

Fifty years of monoclonals: the past, present and future of antibody therapeutics

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Abstract

In 1975, Köhler and Milstein invented hybridoma technology for the generation of murine monoclonal antibodies with predetermined antigen-binding specificity. The transformative impact of monoclonal antibodies is demonstrated by their ubiquitous use as biomedical research reagents and the worldwide approval of at least 212 antibody therapeutics with tens of millions of patients treated to date. Advances in antibody technologies, such as humanization and robust methods for human antibody generation, mitigated the major limitations of murine antibodies as therapeutics. These technologies, combined with progress in biomanufacturing, helped to launch this modern era of antibody therapeutics. Beyond IgG, antibody therapeutics have blossomed into multiple alternative formats, including bispecific antibodies and antibody–drug conjugates. Additionally, antibody fragments have been developed as stand-alone therapeutics and to target cell therapies, notably chimeric antigen receptor T cells. These advances in antibody technologies, plus innovation enabling subcutaneous delivery, have improved the therapeutic benefits and convenience of antibody treatment for many patients. This concept is illustrated here by multiple generations of antibody therapeutics for human epidermal growth factor receptor 2 (HER2)⁺ cancers and B cell-targeted therapies for haematological cancers and immunological diseases. Finally, we opine briefly on some of the many promising future directions with antibody therapeutics, including the application of artificial intelligence for antibody identification and multi-parameter optimization.

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Introduction

In 1975, Georges Köhler and César Milstein¹ published their ground-breaking invention of hybridoma technology for the generation of monoclonal antibodies in *Nature*. Mice were immunized with a target antigen of interest emulsified with an adjuvant to stimulate an immune response (Fig. 1a). B cells isolated from the immunized mice were then fused with myeloma cells to create immortalized cells known as ‘hybridomas’. These hybridoma cells were then screened to identify those that secreted antibodies that selectively bound to the target antigen¹. A defining feature of hybridoma technology is that it provides a perpetual source of antibodies that bind to a target antigen commonly, albeit not invariably², with exquisite selectivity and minimal cross-reactivity to unrelated antigens. Within a few years monoclonal antibodies derived from immunized mice and rats became widely adopted as specific binding reagents of choice for biomedical research and nowadays are ubiquitously used for this purpose. Beyond research reagents, hybridoma was a foundational technology that ultimately led to the current era of antibody therapeutics. The vast potential of hybridoma technology was quickly realized and led to the 1984 Nobel Prize for Physiology or Medicine being awarded to Georges Köhler and César Milstein “for the discovery of the principle for production of monoclonal antibodies”³. This Nobel Prize was shared with Niels Jerne “for theories concerning the specificity in development and control of the immune system”³. More broadly, at least six Nobel Prizes and six Lasker Awards have been made for ground-breaking contributions related to antibodies and antibody therapeutics (Supplementary Table 1).

The transformative impact of hybridoma technology over the past 50 years is hard to overstate. Indeed, hybridoma provided a foundation that has led to at least 212 approved antibody therapeutics to date that have benefited millions of patients (see [Antibody Society](#) website, a meticulously curated and up-to-date tabulation of approved and registrational stage antibody therapeutics). Additionally, the term ‘monoclonal antibody’ or ‘monoclonal antibodies’ is cited in >348,000 scientific publications (PubMed) and in the claims of >15,800 issued US patents. In this Review we celebrate the 50th anniversary of hybridoma technology¹ by discussing the past, present and future of antibody therapeutics.

From a past perspective, key advances beyond hybridoma that contributed to the rise of antibodies as therapeutics are highlighted (Fig. 1b). These include antibody engineering methods, such as chimerization^{4,5} and humanization^{6–8}, and robust technologies for human antibody discovery that largely overcame the substantial shortcomings of murine antibodies as therapeutics (Box 1). Additionally, antibody engineering has been widely used to improve existing properties of antibodies, including modulation of Fc-mediated functions⁹, as well as endowing antibodies with brand new properties, such as bispecificity^{9,10}. Advances in biomanufacturing were also crucial to the clinical development and commercialization of antibodies (Box 2). The landscape of antibody therapeutics has evolved from primarily IgG to numerous additional formats, including bispecific antibodies, antibody–drug conjugates (ADCs) and chimeric antigen receptor (CAR) T cells (Fig. 2a).

Transitioning from the past to present antibody therapeutics, we illustrate the power of reverse translation. Specifically, insights from the clinic in combination with advances in antibody technologies have been used to guide the design of antibody therapeutics that provide greater clinical efficacy, improved patient safety and/or greater convenience to patients. This is exemplified here by reviewing multiple generations of antibody therapeutics targeting B cells – often with

the same target antigen – for haematological malignancies, as well as for inflammatory and autoimmune diseases (Fig. 3). Clinical progress is further illustrated through several generations of antibody therapeutics targeting the human epidermal growth factor receptor 2 (HER2) proto-oncogene in oncology, especially breast cancer (Fig. 4). Major routes of antibody administration are discussed (Fig. 2d), including the reasons that subcutaneous (s.c.) injection is often preferred over intravenous (i.v.) infusion and complementary strategies to enable s.c. delivery (Table 1). Transition from i.v. delivery to s.c. delivery has already been enabled for some antibody therapeutics, including several that target B cells (Fig. 3) or HER2⁺ cancers (Fig. 4).

Segueing from the present to the future, we selected a few areas of innovation with the potential for high patient benefit with next-generation antibody therapeutics, including artificial intelligence for antibody discovery and multi-parameter optimization. Antibody engineering advances discussed include conditional antigen binding for improved anticancer antibodies¹¹, receptor agonism and exploration of immunoglobulin isotypes beyond IgG. One crucial ongoing challenge in the field is minimization of the risk of immunogenicity in patients for highly engineered antibodies¹². As for important antibody delivery challenges, we focus on oral administration and enhancing uptake into the central nervous system. More comprehensive coverage of the future of antibody therapeutics can be found elsewhere¹⁰, including targeting of previously undruggable targets¹³.

Key steps enabling the rise of antibodies as therapeutics

The clinical development time required for innovative therapeutics varies greatly, with a median of approximately 8 years¹⁴. Eleven years elapsed after the invention of hybridoma technology until the approval of the first antibody, muromonab-CD3, by the FDA. Muromonab-CD3 targets CD3 and was approved in 1986 for treatment of acute kidney transplant rejection (Fig. 2a). However, antibodies only started to take off as a major new therapeutic modality in the late 1990s (Fig. 2a). This delay reflects the multiple diverse problems that had to be solved for antibodies to be developed as therapeutics (Fig. 1b). First, basic research was required to investigate the pathobiology of human diseases to identify relevant targets for therapeutic intervention with monoclonal antibodies. Preclinical development followed to assess the safety and efficacy of these antibodies. Early clinical testing of murine antibodies demonstrated very limited efficacy and revealed substantial shortcomings for human therapy: immunogenicity, short plasma half-life and inefficient effector functions (Box 1). Reverse translation of these insights from the clinic led to the advent of technologies that had the potential to mitigate the limitations of murine antibodies. In due course, murine antibodies redesigned through chimerization^{4,5} or humanization^{6–8} or, later, human antibodies (Box 1), were evaluated for their pharmacokinetics, safety and efficacy in clinical trials. The first chimeric antibody, abciximab, targets glycoprotein IIb/IIIa (also known as integrin α IIb β III), and was approved in 1994 to prevent cardiac ischaemic complications in patients undergoing percutaneous coronary intervention¹⁵ (Fig. 2c). The first humanized antibody, daclizumab, targets CD25 and was approved in 1997 in combination with cyclosporin and corticosteroids for prevention of organ rejection in renal transplant recipients¹⁶ (Fig. 2c). The first human IgG antibody, adalimumab, targets tumour necrosis factor (TNF) and was initially approved in 2002 for rheumatoid arthritis¹⁷ (Fig. 2c). Adalimumab has been one of the most clinically and commercially successful antibody therapeutics to date, with FDA approvals in at least nine different chronic inflammatory

a Hybridoma technology

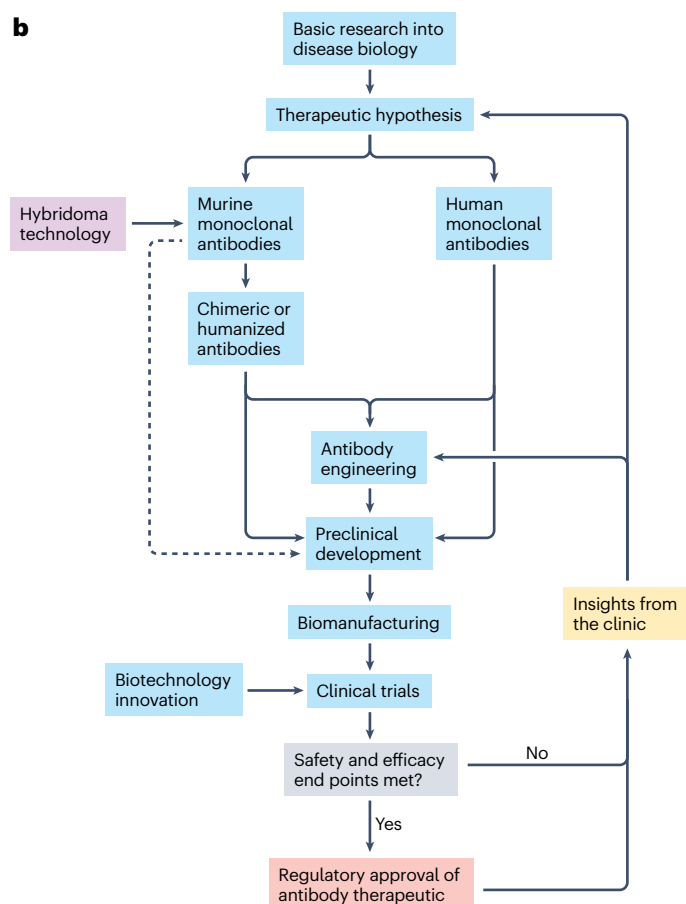
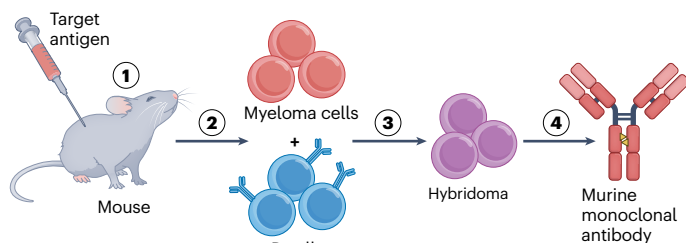


Fig. 1 | From hybridoma technology to antibody therapeutics. **a**, The principal steps in the generation of monoclonal antibodies from hybridoma technology include: (1) immunization of mice or rats with a target antigen of interest plus adjuvant; (2) elicitation of B cells expressing target-specific B cell receptors on their surface; (3) fusion of isolated B cells with myeloma cells to create immortalized cells known as hybridomas; and (4) screening to identify hybridomas secreting monoclonal antibodies that bind selectively to the target antigen – sheep red blood cells in the first published example¹. **b**, Key steps from the advent of hybridoma technology¹ and basic research into disease biology through preclinical development and clinical trials to approval of antibody therapeutics. Fewer than ten murine monoclonal antibodies (excluding use for at least six chimeric antigen receptor (CAR) T cells) have ever been directly developed as **approved therapeutics** (dashed line). Early clinical trials with murine antibodies revealed several crucial limitations as therapeutics, which were largely addressed by the redesign of murine antibodies by chimerization^{4,5} or humanization^{6–8} or alternatively by the use of human antibodies (Box 1). Antibody engineering, for example, to modulate Fc functions⁹, is often used to enhance the clinical potential of antibodies¹⁰. Biotechnology innovation, including in biomanufacturing (Box 2), was also crucial to the emergence of antibody therapeutics. Insights from both successful and unsuccessful clinical trials have been used to develop improved antibody therapeutics as illustrated in this Review for targeting of B cells in oncology, autoimmunity and inflammation (Fig. 3) and human epidermal growth factor receptor 2 (HER2)⁺ tumour cells in oncology (Fig. 4).

(48%), or in autoimmunity and inflammatory diseases (25%) (Fig. 2b). However, a growing number of antibodies have been approved for use in other therapeutic areas, including infectious diseases, haematology, neurology, metabolic diseases, ophthalmology and musculoskeletal diseases (Antibody Society data and ref. 10) (Fig. 2b). Very few approved antibody therapeutics are murine (4%, excluding use in CAR T cells) (Fig. 2c). Indeed, most antibody therapeutics are humanized (51%) or chimeric (8%) versions of murine antibodies. However, a growing number of antibody therapeutics are human (34%) mainly from transgenic mice, but additionally from other sources, including phage display libraries (for example, adalimumab, belimumab, atezolizumab, avelumab and necitumumab), yeast display libraries (for example, tafolcicimab) and human B cells (for example, the following anti-severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) antibodies: amubarvimab, romlusevimab, tixagevimab, cilgavimab, bamlanivimab and etesevimab).

Almost all current antibody therapeutics are approved for delivery either by i.v. infusion (~63%), s.c. injection (~25%) or both i.v. and s.c. (8%) (Fig. 2d). Intravitreal injection is used for a few antibody therapeutics for the treatment of age-related macular degeneration, for example, ranibizumab and brolucizumab-dblI, which target vascular endothelial growth factor (VEGF) and the bispecific antibody faricimab-svoa (targets VEGF and angiopoietin 2). Additional, but rarely used antibody delivery routes include intramuscular injection, for example, palivizumab for respiratory syncytial virus infections and tixagevimab plus cilgavimab for coronavirus disease 2019 (COVID-19) and intraperitoneal infusion for catumaxomab for malignant ascites.

Antibody therapeutics have been dominated from the outset by monospecific IgG, which represent ~74% of currently approved antibodies (see Antibody Society data) (Fig. 2a). However, the past decade has witnessed rapid growth of alternative antibody formats (Fig. 2a), particularly ADCs²¹ (Fig. 2e) and bispecific antibodies^{22,23} (Fig. 2f). Additional approved formats include antibody fragments²⁴, antibody fusion proteins²⁵, separately formulated antibody combinations and

indications¹⁸ and peak year sales exceeding US \$20 billion¹⁹. Antibodies in their prevalent IgG format are large (~150 kDa) and complex glycoproteins with two heavy chains and two light chains (Fig. 3). Much innovation in biotechnology, including biomanufacturing, was needed to produce large quantities (kilograms to multi-tonne) of highly purified and stably formulated antibody therapeutics for clinical trials and commercialization (Box 2).

Evolution of the antibody therapeutics landscape

The landscape of approved antibody therapeutics has grown to at least 212 as of May 2025 (see Antibody Society data) (Fig. 2a) and is expanding rapidly with 18 or more new approvals per year since 2022 (ref. 20). Most antibody therapeutics are approved for indications in oncology

Box 1 | Mitigating the clinical limitations of murine monoclonal antibodies

The high initial expectations of the therapeutic potential of murine antibodies were quickly dashed in the 1980s by a paucity of clinical benefit^{239,240} and the identification of three major and common shortcomings. First, murine antibodies are highly immunogenic foreign proteins, and in most patients they elicit anti-drug antibodies that can compromise antigen binding, pharmacokinetics, safety and/or efficacy^{241,242}. Secondly, murine antibodies are typically less efficient than their human counterparts in supporting effector functions because of numerous differences between murine and human immune systems²⁴³. Thirdly, murine antibodies typically have a plasma half-life of only 1–2 days²⁴⁴ in patients — much shorter than the ~21 day half-life for human IgG1 (ref. 245). A priori, this short half-life is anticipated to limit target exposure and thereby efficacy of murine antibodies. This pharmacokinetic difference is now known to reflect the fact that murine IgG molecules bind much less efficiently to the human recycling receptor, neonatal Fc receptor (FcRn), than do their human IgG counterparts²⁴⁶.

Multiple different technologies have been invented over the past 40 years to mitigate the known limitations of murine antibodies. For example, in chimerization of a murine antibody the antigen-binding variable domains are incorporated into a human antibody^{4,5}. In humanization, the six antigen-binding loops — complementarity-determining regions — within the variable domains are commonly grafted from a murine antibody into a human antibody^{6–8}. A few additional murine residues, such as Vernier zone²⁴⁷, are often also transferred to maintain antigen-binding affinity. Additionally, many robust routes to human antibodies have been developed since the 1990s as alternatives to engineering murine antibodies. Technologies for human antibody discovery include phage²⁴⁸, yeast²⁴⁹ and ribosome²⁵⁰ display libraries; transgenic mice^{251,252}, rats^{253,254}, rabbits²⁵⁵ and chickens²⁵⁶; and high-throughput antibody cloning from human B cells²⁵⁷. Combinations of different technologies are increasingly being used for antibody discovery, for example, ribosome display of antibody fragment libraries in conjunction with deep sequencing²⁵⁰.

coformulated antibody mixtures²⁶. Another innovation is the use of antibody fragments displayed on cell surfaces to target cellular therapeutics. Indeed, at least seven different CAR T cells have been approved by the FDA for the treatment of B cell cancers (Fig. 3), including six or more targeted with single-chain (sc) Fv fragments²⁷ and one targeted with a biparatopic single-domain (VHH) antibody (ciltacabtagene autoleucel). Additional diversity of antibody formats is evident from preclinical and early clinical research, including but not limited to antibody prodrugs, multispecific antibodies and ADCs with various payloads, including nucleic acids and immunomodulators^{10,13}. One recent globalization trend is the growing number of antibody therapeutics for which the first and sometimes sole approval to date was outside the USA and Europe (see [Antibody Society](#) data). For example, two of the antibody therapeutics targeting HER2⁺ cancers (Fig. 4) received their first approval in China, namely, the ADC disitamab vedotin for gastric cancer and inetetamab, an IgG1 with enhanced effector functions for metastatic breast cancer (see [Antibody Society](#) data).

Antibody–drug conjugates

The concept of arming antibodies through conjugation to protein toxins, known as immunotoxins, originated in the early 1970s²⁸. Pioneering work involved chemical conjugation of antibodies to potent protein payloads such as diphtheria toxin to induce cytotoxicity against malignant cells²⁸. This research laid the groundwork for subsequent developments in ADCs a few years later²¹. The core underlying concept of ADCs is to use the targeting specificity of antibodies to enhance the therapeutic index of highly cytotoxic small molecules by increasing on-tumour activity and reducing unwanted off-target effects. The real-world experience with ADCs in patients is much more complex^{21,29}. Indeed, clinical trials with many different radiolabelled antibodies revealed that only a tiny percentage of administered antibodies accrues in the tumour ($\leq 0.1\%$ injected dose per gram of tumour) with most being catabolized elsewhere in the body³⁰; presumably also the case with ADCs³¹. Nevertheless, ADCs radically extend the plasma half-life of small-molecule drugs, provide some enhancement of their therapeutic index and allow some preferential localization to tumour cells. ADC binding to cell surface antigens induces their internalization, typically through the endolysosomal pathway, and subsequently releases their cytotoxic payloads in lysosomes. The payloads then need to exit lysosomes by passive diffusion or active transport to reach the sites of action, such as microtubules in the cytosol, or to cross the nuclear membrane to act on DNA in the nucleus. These cellular uptake mechanisms allow ADCs to deliver cell-impermeable payloads, such as monomethyl auristatin F³², in addition to cell-permeable payloads, such as the closely related monomethyl auristatin E³³. The mechanisms of payload release from the ADC depend upon the design of the linker (cleavable or non-cleavable) and include proteolysis of the antibody and/or linker, reduction of disulfide bonds or hydrolysis of acid-labile bonds as discussed in detail elsewhere²¹. For non-cleavable linkers, the mechanism of drug release is proteolytic catabolism of the entire ADC to release active catabolites.

In 2000, the first ADC was approved by the FDA, namely, gemtuzumab ozogamicin, a humanized anti-CD33 antibody covalently linked to the highly cytotoxic antibiotic, *N*-acetyl calicheamicin, through an acid-labile linker. Gemtuzumab ozogamicin binds to CD33, an antigen expressed in ~90% of acute myeloid leukaemia (AML) cells. Gemtuzumab ozogamicin internalization leads to release of calicheamicin, which induces double-stranded DNA breaks that induce AML cell death. Gemtuzumab ozogamicin was approved for the treatment of relapsed AML³⁴. However, it was withdrawn in 2010 when confirmatory trials failed to demonstrate clinical benefit and revealed serious drug-associated adverse events including sometimes fatal hepatotoxicity and veno-occlusive disease. Gemtuzumab ozogamicin was re-approved by the FDA in 2017 with a lower fractionated dosing schedule that improved the safety profile while providing clinical efficacy³⁵.

The approximately 30-year lag between the advent of immunotoxins²⁸ and the approval of the first ADC reflects decades of innovation across a broad range of disciplines to make ADCs beneficial to patients. ADCs are inherently complex therapeutics, including their pharmacology and manufacture, as they combine the intricacies of protein (antibody) and small-molecule (payload and linker) drug development²¹. A plethora of crucial decisions are required in the design of successful ADCs, including the choice of target antigen, antibody, cytotoxic payload, linker and drug-to-antibody ratio (DAR), as well as the conjugation chemistry and sites of drug attachment, as extensively reviewed elsewhere²¹. Commonly used chemistries for

conjugation of linker-payloads to antibodies use surface lysine residues, or alternatively naturally occurring³³ or engineered³⁶ cysteine residues²¹.

The number of approved ADCs – all in oncology to date – has grown rapidly in recent years and now totals at least 16, excluding two radioimmunoconjugates²¹ (Fig. 2e). Beyond calicheamicin, the cytotoxic payloads for approved ADCs exhibit several different mechanisms of action that ultimately result in cell death²¹ (Fig. 2e). For example, auristatins and maytansinoids inhibit tubulin polymerization, leading to mitotic arrest and apoptotic cell death. Pyrrolbenzodiazepine dimers bind to the DNA minor groove, resulting in DNA alkylation, DNA cross-linking and cell death. Camptothecins inhibit topoisomerase I, which leads to DNA damage, cell cycle arrest and cell death. Future trends in ADCs include the exploration of alternative small-molecule payloads to increase their therapeutic index or enhance their drug-like properties with applications that extend beyond oncology²¹. Payloads being investigated include antibiotics³⁷, immunomodulators, protein degraders^{38,39}, molecular glues³⁹ and nucleic acids.

Bispecific antibodies

In 1960, the concept of bispecific antibodies was first demonstrated by Nisonoff et al.⁴⁰ using disulfide conjugation of Fab fragments obtained by pepsin cleavage of rabbit IgG. In 2009 the first bispecific antibody, catumaxomab (which targets epithelial cell adhesion molecule (EpCAM) and CD3) was approved by the EMA for the treatment of malignant ascites⁴¹. Catumaxomab was the first-in-class bispecific antibody designed to recruit T cells to kill tumour cells; such drugs are now commonly known as T cell engagers (TCEs). Catumaxomab was removed from the market for commercial reasons but then re-authorized in Europe in 2025. The nearly 50-year lag between the advent of bispecific antibodies⁴⁰ and the approval of catumaxomab reflects decades of enabling innovation, including the advent of hybridoma¹ and hybrid hybridoma⁴² technologies for the generation of monospecific and bispecific antibodies, respectively. Additionally, basic research provided the conceptual breakthrough of using bispecific antibodies to direct the killing of tumour cells by T cells⁴³. The advent of efficient biomanufacturing methods for bispecific antibodies was also crucial to the clinical development and commercialization of these complex and commonly multi-chain proteins⁴⁴.

The number of approved bispecific antibodies has increased rapidly, particularly since 2020, and now totals at least 20 (Fig. 2f). These are approved primarily for use in oncology²³ and include 12 or more TCEs. Bispecific antibodies are also approved for disease indications in ophthalmology (faricimab-svoa, which targets VEGF and angiopoietin 2), haematology (emicizumab-kxwh, which targets factor IXa and factor X) and rheumatology (ozoralizumab). For most bispecific antibodies, the two binding specificities are directly related to the mechanism of action. In contrast, the trivalent VHH-based bispecific antibody fragment ozoralizumab is bivalent for binding to TNF to support the mechanism of action and monovalent for binding to albumin to extend the pharmacokinetic half-life.

The modular architecture of immunoglobulins has been used by creative antibody engineers to design an impressively diverse array of more than 100 different bispecific antibody formats⁴⁵, to enable exquisite customization of properties for specific clinical settings. Tunable features of bispecific antibodies include the number of binding sites for each antigen (valency), their binding affinity and spatial geometry for binding the two antigens, molecular weight, pharmacokinetics and optional effector functions⁴⁵. One useful classification scheme for bispecific antibodies is their valency for each antigen-binding

specificity, for example, 1+1, 2+1 and 2+2 formats^{45,46} (Figs. 2f, 3 and 4). A major protein engineering challenge with such bispecific antibodies, now largely solved, has been to facilitate the preferential pairing of cognate heavy and light chains and to promote heavy chain heterodimerization for Fc-containing bispecific antibodies⁴⁵. Antibody technologies used in the efficient molecular assembly of currently approved bispecific antibodies include BiTE⁴⁷, knobs-into-holes^{48,49}, CrossMab^{50,51}, common light chains⁵², DuoBody⁵³ and Azymetric⁵⁴. Beyond bispecific antibodies many trispecific and multispecific antibodies have advanced to clinical development^{22,55}, although they have not yet reached approval by health authorities.

Improved patient outcomes from advances in antibody technologies

Novel antibody engineering technologies in conjunction with delivery innovation have enabled the development of antibody therapeutics that provide greater benefit to patients. Progressive improvement in

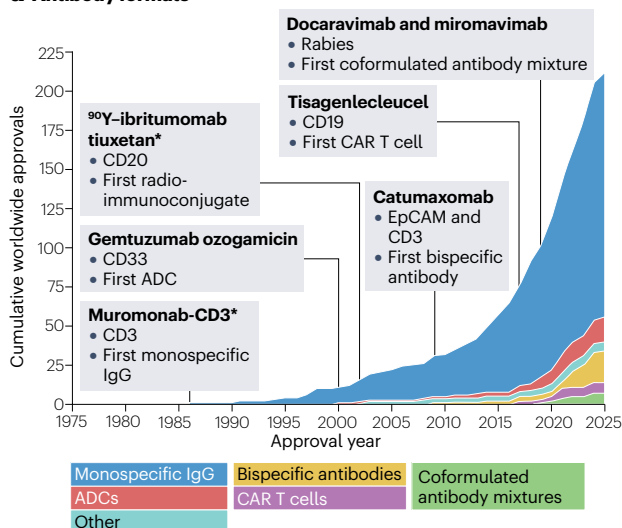
Box 2 | Biomanufacturing of antibodies

The emergence of antibodies as therapeutics was enabled in part by progress in biotechnology over nearly 50 years, including the large-scale (kilogram to multi-tonne) recombinant production of high-quality and meticulously characterized recombinant proteins^{258,259}. Almost all antibody therapeutics are manufactured in either Chinese hamster ovary (CHO, ~73%) or myeloma (~22%) cell lines (see [Antibody Society](#) data). Reliable mainstays of antibody manufacturing for decades have been large volume (commonly 2,000–25,000 l) batch cell culture in stainless steel bioreactors and chromatographic purification using protein A and other resins^{258,260}. Antibody titres in CHO cell cultures have increased from <1–2 g l⁻¹ in the late 1990s to the 5–8 g l⁻¹ that is routine today²⁵⁸ with titres of >10 g l⁻¹ achieved for some antibodies²⁶¹. These large titre increases reflect substantial improvements in CHO cell line development and culture, for example, site-specific integration of expression cassettes into the CHO host genome²⁶¹, host cell line engineering²⁶², media optimization²⁶³, fed batch²⁶⁴ and high cell density culture, including by perfusion²⁶⁵.

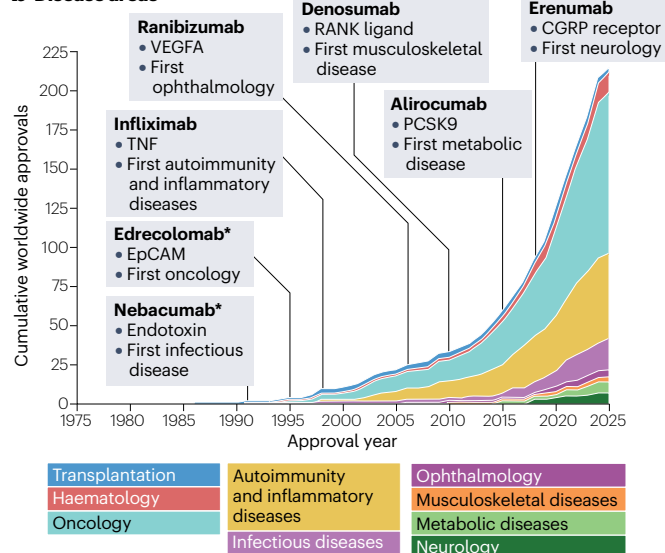
A major area of innovation in antibody biomanufacturing is the advent of single-use equipment, particularly for cell culture volumes of ≤2,000 l. Benefits of single-use technologies include lower contamination risk, reduced cleaning validation and flexible capacity²⁶⁶. Challenges of single-use biomanufacturing technologies include scaling above 2,000 l, potential contamination from leachable and extractable chemicals²⁶⁷ and waste generation²⁶⁶. Sustainability benefits from single-use biomanufacturing are large savings in water and energy needed to clean and sterilize stainless steel facilities. Substantial progress in antibody biomanufacture has also been made in continuous processing, leading to faster production times and reduced cost of goods²⁶⁰. Multiple obstacles to broader adoption of continuous processing include the need for greater automation, real-time process monitoring, control systems to optimize cell growth and more sophisticated process analytical technologies²⁶⁶. The landscape of antibody biomanufacturing continues to evolve with traditional batch processes operating alongside emerging single-use technologies combined with continuous processing^{268,269}.

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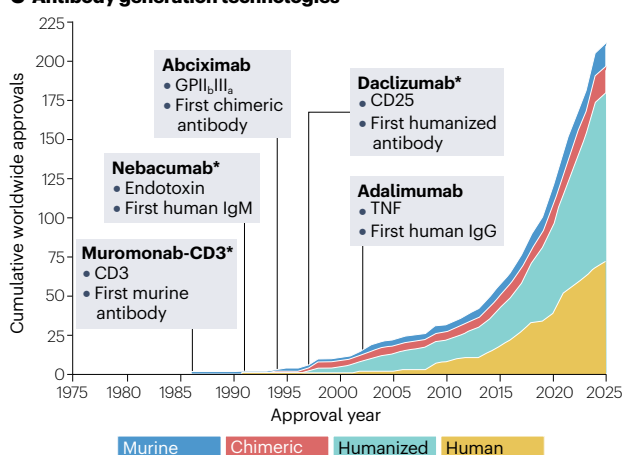
a Antibody formats



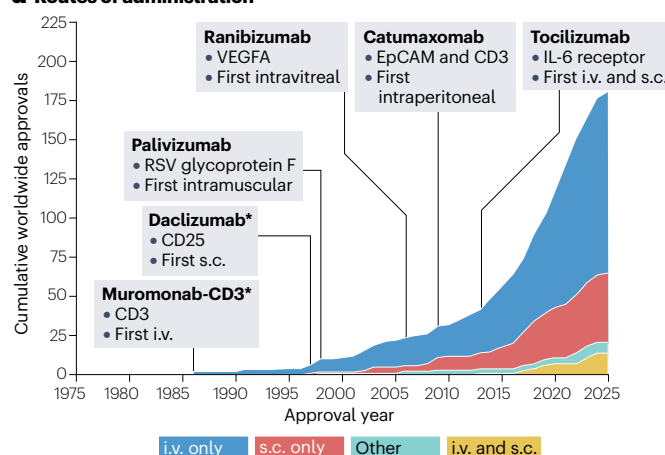
b Disease areas



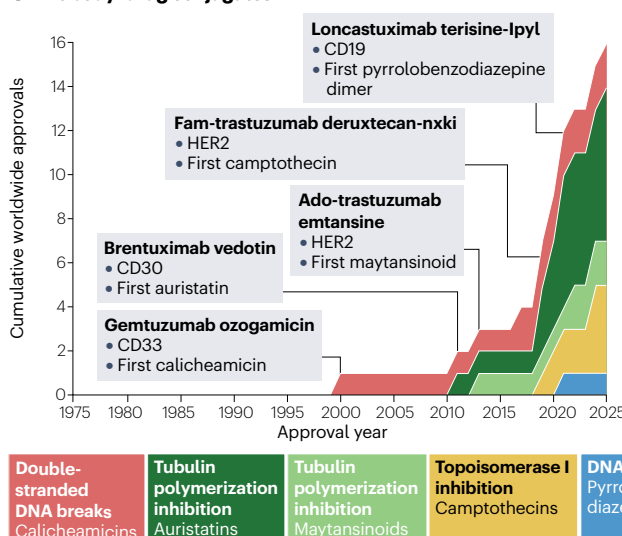
c Antibody generation technologies



d Routes of administration



e Antibody-drug conjugates



f Bispecific antibodies

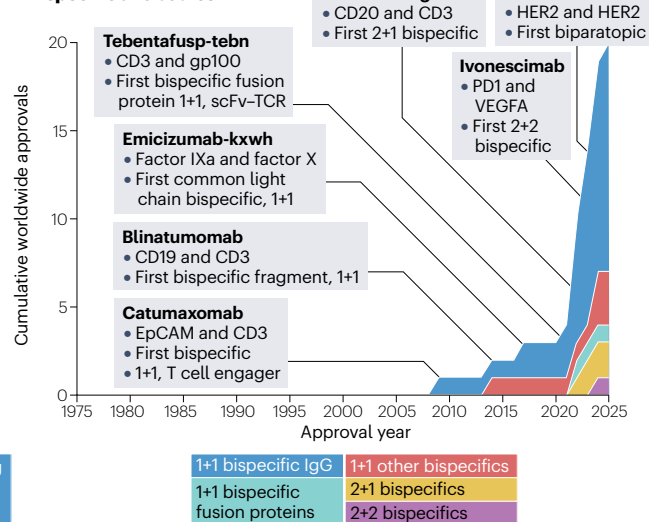


Fig. 2 | The rise of antibodies as therapeutics. Cumulative worldwide regulatory approvals up until May 2025 from the [Antibody Society](#) for all antibodies organized by: formats (part **a**); disease areas (part **b**); generation technologies (part **c**); and routes of administration (part **d**). Also shown are antibody–drug conjugates (ADCs) excluding radioimmunoconjugates (part **e**) and bispecific antibodies (part **f**). Notable first antibody approvals are shown along with their cognate antigens (see [Antibody Society](#) data). Here, antibodies are defined broadly as protein therapeutics with one or more antigen-binding immunoglobulin variable domains and typically additional domains and/or components. Also included are cell-based therapies that use surface display of single-chain Fv fragment (scFv) or VHH antibody fragments for targeting of chimeric antigen receptor (CAR) T cells. FDA approval data for CAR T cells were obtained from Mulvey et al.²³⁸ plus search of the FDA website. The ‘other’ category in part **a** includes monospecific antibody fragments, radioimmunoconjugates, immunocytokines and antibody fusion proteins. Not included are more than 100 approved biosimilar antibodies, coformulated

antibodies that were previously approved separately, such as trastuzumab plus pertuzumab or trastuzumab and hyaluronidase, and Fc-fusion proteins that lack immunoglobulin variable domains. At least nine antibody therapeutics have been withdrawn from the market for commercial or regulatory reasons, including some of the highlighted firsts (*) (see [Antibody Society](#) data). Routes of administration for antibody therapeutics were compiled by us from available prescribing information from the FDA and the EMA. A few antibodies that were approved first for intravenous (i.v.) and later subcutaneous (s.c.) delivery are moved between categories in the plot. For example, tocilizumab was approved first for i.v. infusion (2005, Japan) and then later for s.c. injection (2013). CGRP, calcitonin gene-related peptide; EpCAM, epithelial cell adhesion molecule; HER2, human epidermal growth factor receptor 2; PCSK9, proprotein convertase subtilisin/kexin type 9; RANK, receptor activator of nuclear factor- κ B; RSV, respiratory syncytial virus; TNF, tumour necrosis factor; VEGFA, vascular endothelial growth factor A.

clinical efficacy, safety and convenience is exemplified below through multiple generations of antibody therapeutics targeting B cells or the proto-oncogene HER2.

Monospecific IgG antibodies for B cell targeting in cancer

B cell malignancies are particularly tractable indications for antibody therapies. This reflects that there are multiple surface targets that are expressed predominantly on B cells and B cell malignancies, for example, CD20, CD19, CD79 and CD22, and depletion of normal B cells does not lead to dose-limiting toxicities. Before 1997, the mainstay for treatment of non-Hodgkin lymphoma (NHL) was combination chemotherapy: cyclophosphamide, doxorubicin hydrochloride, vincristine sulfate and prednisone. In 1997, rituximab became the first anti-CD20 antibody to be FDA-approved for the treatment of NHL⁵⁶. Rituximab is a chimeric antibody with variable domains derived from the mouse anti-human CD20 antibody, IDEC-2B8, plus human κ light chain and IgG1 constant heavy chain domains⁵⁷. CD20 is predominantly expressed on B-lineage cells, appearing first on pre-B cells, and then its expression is extinguished as B cells mature into antibody-secreting plasma cells. Selective targeting of B cell subsets with rituximab preserves most pre-existing antibodies, particularly from long-lived plasma cells that are CD20 negative. In haematological cancers, CD20 is expressed on most malignant B cell leukaemias and lymphomas. Targeting of CD20 with rituximab has revolutionized the treatment of multiple B cell malignancies, notably, follicular lymphoma (FL), diffuse large B cell lymphoma (DLBCL) and chronic lymphocytic leukaemia (CLL)⁵⁶.

Rituximab eliminates CD20⁺ B cells through multiple mechanisms – predominantly involving immune effector functions. These include antibody-dependent cellular cytotoxicity (ADCC), through Fc γ receptor interactions with natural killer cells, macrophages and neutrophils; antibody-mediated cellular phagocytosis (ADCP), also through Fc γ receptors expressed on phagocytes; complement-mediated cytotoxicity (CDC); and direct apoptotic pathways. Two types of anti-CD20 antibody (types I and II) have been proposed on the basis of differences in their ability to organize CD20 molecules into lipid rafts⁵⁸. In addition, type II anti-CD20 antibodies (obinutuzumab) can directly induce a caspase-independent and lysosome-mediated cell death⁵⁸. However, the biophysical basis for these differences is not well understood. Recent application of high-resolution imaging methods of type I (rituximab and ofatumumab) and type II (obinutuzumab) antibodies demonstrate that both classes are capable of cross-linking and concatenating

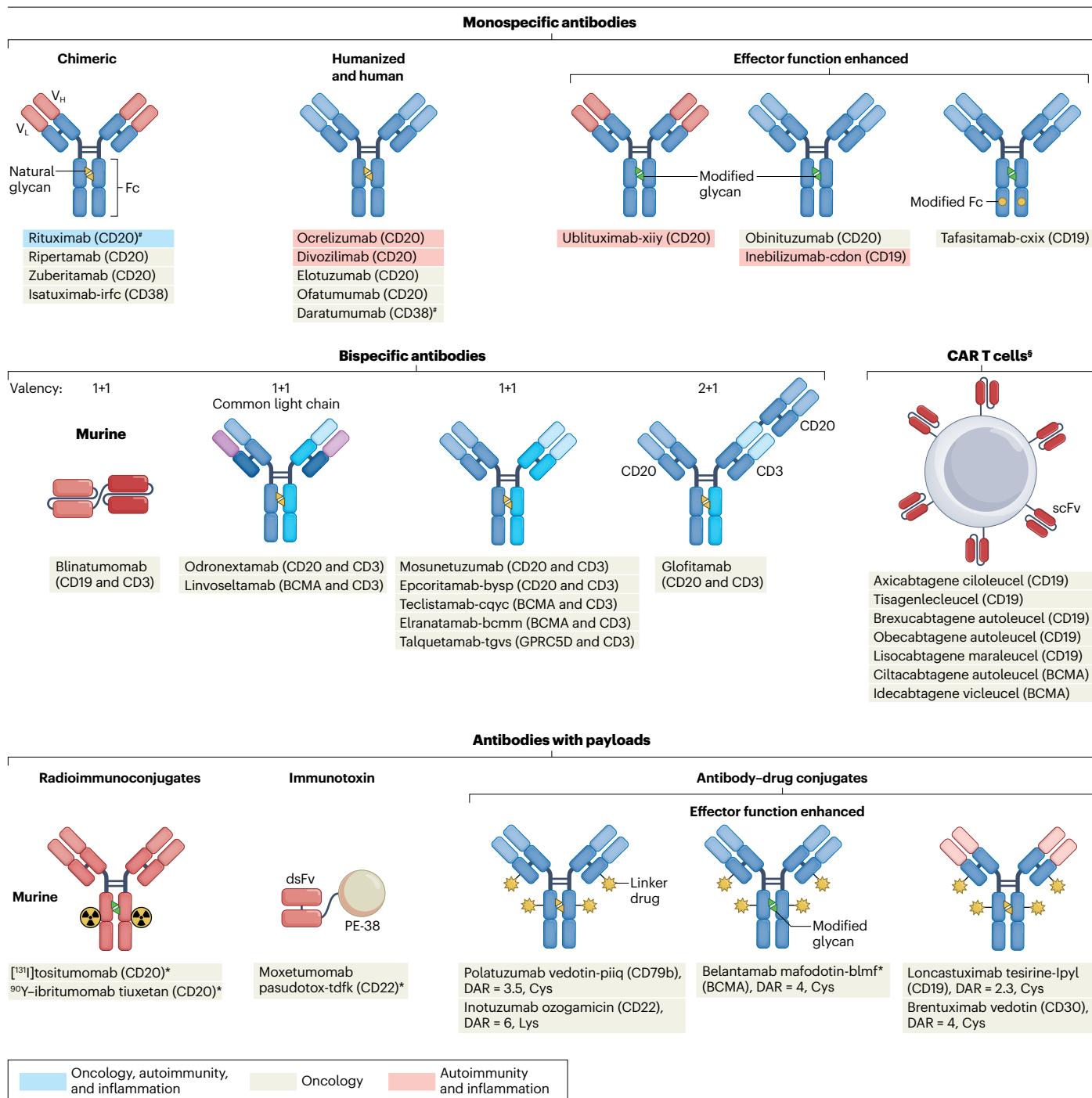
CD20 on the Raji B cell line⁵⁹. Additionally, live-cell imaging demonstrates that rituximab, and to a much lesser degree obinutuzumab, polarizes and accumulates antibody–CD20 complexes on microvilli, accompanied by the formation of long microvilli protrusions. These additional biophysical and cellular insights may provide an opportunity for future generation of antibody therapeutics.

The clinical observation that patients with a higher affinity polymorphism in Fc γ RIIIA (F158) and Fc γ RIIA (H131) have improved responses with rituximab therapy^{60,61} provided a rationale to develop obinutuzumab. Beyond directly inducing B cell cytotoxicity, obinutuzumab is glyco-engineered to enhance ADCC. Specifically, obinutuzumab is manufactured in a Chinese hamster ovary (CHO) cell line overexpressing β 1,4-*N*-acetylglucosaminyltransferase III to increase bisecting *N*-acetylglucosamine (GlcNAc) on the glycan attached to the Fc residue N297. This in turn decreases fucosylation at N297 and thereby enhances binding to Fc γ RIIIA on effector cells⁶². Such glyco-engineering to enhance ADCC and induction of cell death contributes to the benefit observed with obinutuzumab in patients with rituximab-refractory NHL and in the greater benefit for patients with CLL of obinutuzumab in combination with chlorambucil when compared with rituximab with chlorambucil^{63,64}. However, obinutuzumab did not demonstrate superiority in overall survival in untreated indolent NHL⁶⁵ or in DLBCL⁶⁶. These differences in clinical outcome highlight how therapeutic effector mechanisms of action, even to the same target, need to be customized according to the underlying molecular causes of the disease to be beneficial. Obinutuzumab is currently FDA-approved for treatment of CLL and FL for patients who are refractory or relapsed (R/R) after rituximab treatment and for patients with previously untreated stage II bulky, stage III or stage IV FL^{67,68}.

CD19, like CD20, is a targetable B cell lineage marker. CD19 is more broadly expressed than CD20 on the surface of B-lineage cells: pro-B cells, throughout B cell maturation and on most plasma cells. Tafasitamab-cxix is a humanized anti-CD19 IgG1 with two Fc mutations (S239D:I332E) to enhance ADCC and ADCP⁶⁹ that is approved for treatment of patients with R/R DLBCL^{70,71} (NCT05429268) in combination with lenalidomide. The availability of multiple targets, such as CD19 or CD20, also provides patients with better therapeutic options, as disease relapse is often associated with loss of antigen expression⁷².

Beyond monospecific antibodies for B cell targeting in cancer

The ability to create bispecific antibodies has enabled a new therapeutic platform by which immune cells, such as T cells, natural killer cells or



myeloid cells, can be recruited to tumours, particularly haematological malignancies, to effect target cell killing. TCEs are highly potent, with the ability to target tumour cells with low level of surface antigen expression. TCEs have demonstrated their greatest potential to date in oncology targeting antigens that are expressed on haematological tumours, including CD19 or CD20 for B cell cancers plus BCMA and GPRC5D for plasma cell dyscrasias (Fig. 3). Additionally, TCEs show promise for antigens such as DLL3 that are overexpressed on solid tumours but have very little expression in normal tissue, as exemplified

by tarlatamab-dlle (targeting DLL3 and CD3), which is approved for treatment of small-cell lung cancer⁷³.

In 2014, blinatumomab became the first TCE approved by the FDA for treatment of acute lymphoblastic leukaemia (ALL)⁷⁴. Blinatumomab consists of two scFv fragments arranged in a tandem format (BiTE), with one scFv binding to CD19 and the second scFv binding to the T cell receptor (TCR)-associated CD3ε signalling subunit joined by a glycine–serine linker⁷⁵ (Fig. 3). Blinatumomab has a very short plasma half-life (~2 h) reflecting renal clearance resulting from its

Fig. 3 | Antibody therapeutics targeting B cells in disease. Most of these antibodies are approved in oncology with a few approved for autoimmunity and inflammation. Rituximab is approved in both of these major disease areas. Approved murine, chimeric, humanized and human antibody therapeutics targeting B cells are shown, including their cognate antigens. Murine sequences (red) and human sequences (blue) are highlighted with the variable domains for light (V_L) and heavy (V_H) chains shown in lighter shades of the same colours. Enhancement of effector functions has been achieved by glyco or protein engineering modifications to the Fc region. For bispecific antibodies, the valency for each antigen (1+1 or 2+1) is indicated. The bispecific antibodies, odronextamab (targeting CD20 and CD3)⁸¹ and linvoseltamab (targeting B cell

maturation antigen (BCMA) and CD3)⁸⁴, use a common light chain⁵² (purple). Biosimilar antibodies are not included. #Rituximab and daratumumab were first approved for intravenous (i.v.) delivery and then subsequently coformulated with hyaluronidase and approved for subcutaneous (s.c.) delivery. *Antibody therapeutics withdrawn from the market. §Chimeric antigen receptor (CAR) T cells are targeted with single-chain Fv fragments (scFv) as shown, except for ciltacabtagene autoleucel, which is targeted with biparatopic VHH fragments. Antibody–drug conjugates were created by conjugation of linker–payloads to either lysine (Lys) or cysteine (Cys) residues on antibodies as indicated. CDR, complementarity-determining region; DAR, drug-to-antibody ratio; dsFv, disulfide-stabilized Fv fragment.

low molecular weight (~55 kDa) and lack of Fc region. Blinatumomab is administered by continuous i.v. infusion over 4-week cycles⁷⁶. Cytokine release syndrome (CRS) is commonly observed with TCE therapy because of T cell activation to kill the tumour cell. Hence, the short half-life of blinatumomab was advantageous in the early days of TCE development as treatment could be readily halted if patients experienced CRS or other adverse events. This early benefit of short half-life for TCEs was rendered less desirable by alternative treatment innovations: corticosteroids and/or anti-IL-6 receptor antibody for prophylaxis and/or treatment of CRS, step-dosing (smaller first dose with higher subsequent doses) and s.c. dosing. Indeed, the field has now shifted to TCEs with longer plasma half-lives, such as bispecific IgG, and, less commonly, half-life-extended BiTEs, as exemplified by tarlatamab⁷³. This transition to TCEs with longer plasma half-lives eases patient drug delivery burden with i.v. dosing frequencies commonly every 3–4 weeks and s.c. dosing at home possible in some cases⁷⁷.

Four TCEs targeting CD20 have been approved for treatment of B cell lymphomas: mosunetuzumab for R/R FL⁷⁸, glofitamab-gxbm for R/R DLBCL and large B cell lymphoma⁷⁹, epcoritamab-bysp for R/R DLBCL⁸⁰ and odronextamab for R/R FL⁸¹. The TCEs, teclistamab-cqyv⁸², elranatamab-bcmm⁸³ and linvoseltamab⁸⁴, targeting B cell maturation antigen (BCMA) – a crucial survival receptor for plasma cells – have been approved for treatment of multiple myeloma. Also approved for multiple myeloma is the TCE talquetamab-tgvs⁸⁵, which targets GPCR5D, a class C orphan G protein-coupled receptor. Most TCEs (six of seven) approved for targeting B cell malignancies have a 1+1 bispecific IgG format (Fig. 3). In contrast, glofitamab-gxbm has a novel 2+1 CrossMab⁵¹ structure in which an anti-CD3 Fab is genetically fused to the ‘bottom’ one of anti-CD20 Fab and a second anti-CD20 Fab is present on the other arm to provide bivalency for CD20 (Fig. 3). This 2+1 valency for CD20 and CD3 provides superior avidity for CD20 over the standard 1+1 valency that translates into greater potency⁸⁶.

A recent extension of antibody engineering is for cell-based therapeutics, notably CAR T cells. Unlike the antibody therapeutics discussed above, this platform involves a molecularly engineered CAR, consisting of an antibody fragment (scFv or biparatopic VHH fragment) fused to a transmembrane domain and an intracellular domain that typically includes the signalling domains from either CD28 or CD137 (4-1BB) and the CD3ζ chain of the TCR, into a patient’s autologous T cells⁸⁷. These cells are then expanded in vitro and re-infused into patients. The scFv binds to a tumour antigen – CD19 for B cell malignancies and BCMA for multiple myeloma – to induce T cell activation and killing of the tumour. Five CAR T cell therapies targeting CD19 are approved by the FDA for the treatment of various B cell lymphomas: axicabtagene ciloleucel, tisagenlecleucel, brexucabtagene autoleucel, obecabtagene autoleucel and lisocabtagene maraleucel. Additionally,

two CAR T cells targeting BCMA are approved by the FDA for treatment of multiple myeloma: ciltacabtagene autoleucel and idecabtagene vicleucel^{88,89}. Unlike protein-based antibody therapeutics, CAR T cells are living drugs that proliferate and dynamically evolve into other cell states over time⁹⁰. This is an exciting convergence between protein and cellular biotherapeutics about which there is much still to be learnt in the future.

Orthogonally to these antibody and scFv-based platforms, antibody conjugates have also transformed the treatment of haematological malignancies. Radioimmunoconjugates were among the first antibodies with payloads approved for therapeutic use. Ibritumomab tiuxetan conjugated to yttrium-90 and [¹³¹I]tositumomab were both approved by the FDA for treatment of B cell malignancies^{91,92}. However, they have had limited clinical and commercial impact, reflecting their high complexity of manufacture, complicated clinical administration and the development of simpler and more efficacious B cell-targeted therapies.

Since the approval of the first ADC, gemtuzumab ozogamicin (discussed above), five ADCs have been approved for treatment of B cell lymphomas, leukaemias and multiple myeloma (Fig. 3). These ADCs target different antigens with diverse payloads: CD30 (monomethyl auristatin E), CD22 (calicheamicin), CD79b (monomethyl auristatin E), CD19 (pyrrolobenzodiazepine dimer) and BCMA (monomethyl auristatin F). Inotuzumab ozogamicin – the second ADC armed with calicheamicin – targets CD22, which is expressed on normal and malignant B cells, as well as on plasmacytoid dendritic cells, and is approved for treatment of R/R CD22-positive B cell precursor ALL⁹³. Polatuzumab vedotin-piiq is an ADC consisting of a humanized anti-CD79b antibody conjugated via a protease-cleavable linker to monomethyl auristatin E, a microtubule inhibitor. CD79b is a transmembrane signalling subunit of the B cell antigen receptor and is rapidly internalized following binding at the cell surface. Polatuzumab vedotin-piiq is approved for treatment of newly diagnosed large B cell lymphoma in combination with rituximab, cyclophosphamide, doxorubicin and prednisone (pola-R-CHP)⁹⁴. Loncastuximab tesirine-lpyl is an ADC consisting of a humanized anti-CD19 antibody conjugated to the pyrrolobenzodiazepine dimer, SG3199 (ref. 95). In contrast to auristatins, pyrrolobenzodiazepine dimers are inter-strand minor groove DNA cross-linkers that have the advantage of not distorting native DNA structures, thereby evading DNA repair mechanisms. Loncastuximab tesirine-lpyl is approved for treatment of R/R DLBCL⁹⁵. Belantamab mafodotin-blmf is an ADC consisting of a humanized antibody targeting BCMA, glyco-engineered (afucosylated) Fc for enhanced ADCC, conjugated to monomethyl auristatin F via a non-cleavable maleimidocaproyl linker⁹⁶. Belantamab mafodotin-blmf was approved for R/R multiple myeloma but withdrawn from the market when a confirmatory phase III trial,

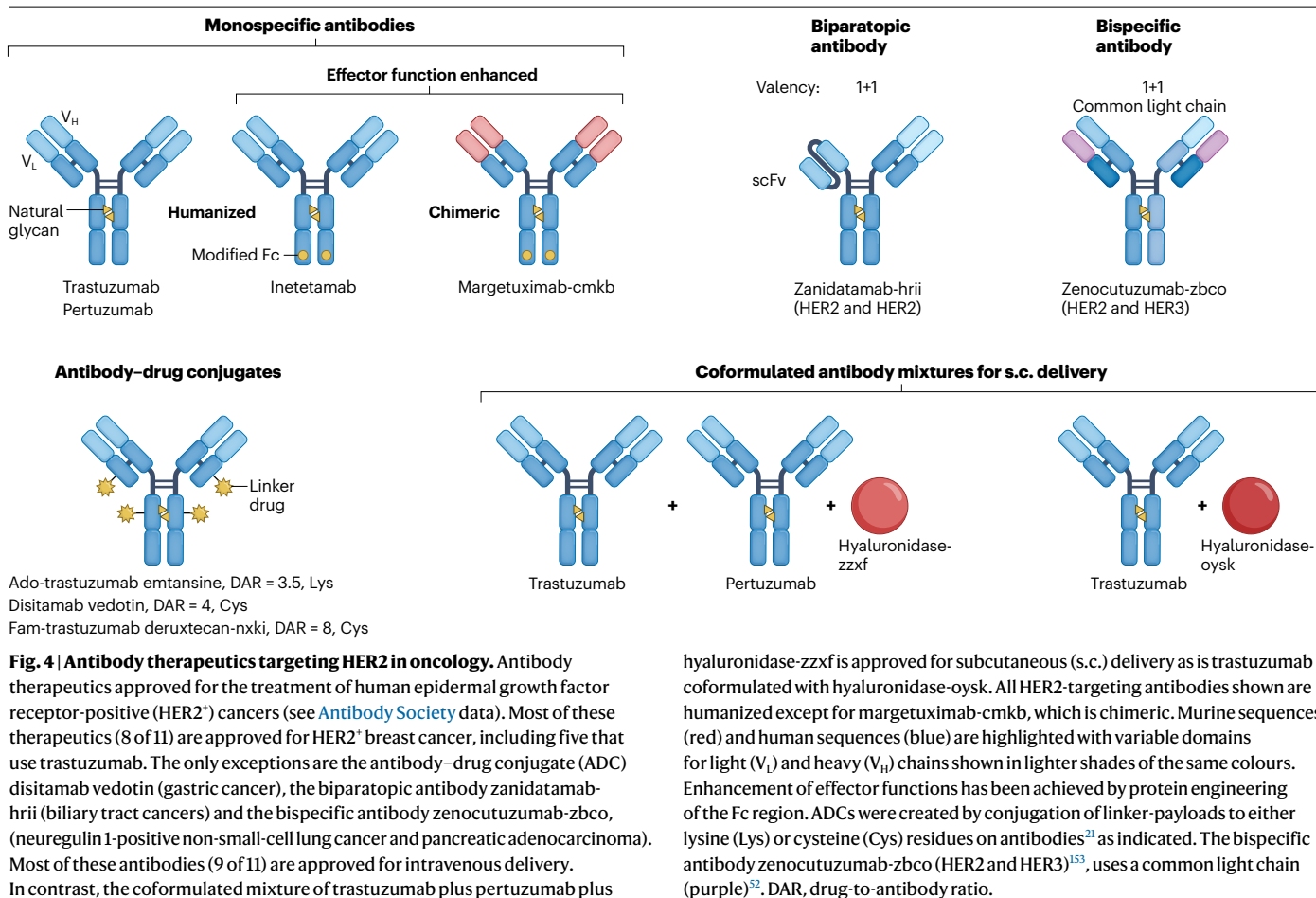


Fig. 4 | Antibody therapeutics targeting HER2 in oncology. Antibody therapeutics approved for the treatment of human epidermal growth factor receptor-positive (HER2⁺) cancers (see [Antibody Society](#) data). Most of these therapeutics (8 of 11) are approved for HER2⁺ breast cancer, including five that use trastuzumab. The only exceptions are the antibody–drug conjugate (ADC) disitamab vedotin (gastric cancer), the biparatopic antibody zanidatamab-hrII (biliary tract cancers) and the bispecific antibody zenocutuzumab-zbco, (neuregulin 1-positive non-small-cell lung cancer and pancreatic adenocarcinoma). Most of these antibodies (9 of 11) are approved for intravenous delivery. In contrast, the coformulated mixture of trastuzumab plus pertuzumab plus

hyaluronidase-zzxf is approved for subcutaneous (s.c.) delivery as is trastuzumab coformulated with hyaluronidase-oysk. All HER2-targeting antibodies shown are humanized except for margetuximab-cmkb, which is chimeric. Murine sequences (red) and human sequences (blue) are highlighted with variable domains for light (V_L) and heavy (V_H) chains shown in lighter shades of the same colours. Enhancement of effector functions has been achieved by protein engineering of the Fc region. ADCs were created by conjugation of linker-payloads to either lysine (Lys) or cysteine (Cys) residues on antibodies²¹ as indicated. The bispecific antibody zenocutuzumab-zbco (HER2 and HER3)¹⁵³, uses a common light chain (purple)⁵². DAR, drug-to-antibody ratio.

DREAMM-3, failed to show a progression-free survival advantage over the comparator arm (pomalidomide plus low-dose dexamethasone)⁹⁷.

Beyond small-molecule payloads, moxetumomab pasudotox-tdfk is a recombinant immunotoxin consisting of a disulfide-stabilized Fv fragment targeting CD22 that is fused to a 38 kDa fragment of *Pseudomonas* exotoxin PE38 (ref. 98). After internalization into CD22-bearing cells, PE38 induces caspase-mediated apoptosis through mitochondrial damage and blockade of translational elongation factor 2. In 2018 moxetumomab pasudotox-tdfk was approved for treatment of hairy cell leukaemia but was subsequently discontinued for business reasons.

In summary, major advances in antibody design over the past 30 years now provide patients with B cell malignancies with multiple therapeutic options, notably monospecific antibodies, bispecific antibodies such as TCEs, as well as ADCs and CAR T cells (Fig. 3). The target antigens on B cells described here are lineage markers rather than oncogenic drivers. Consequently, loss of antigen expression and upregulation of drug transporters, as well as other resistance mechanisms, often cause disease relapse⁹⁹. However, the availability of distinct therapeutic antigens in conjunction with diverse antibody therapeutics with multiple mechanisms of action provide many options to extend the lives of patients and potentially cure them of their disease, perhaps even without cytotoxic chemotherapy.

B cell targeting in immune-mediated diseases

One of the first randomized controlled studies demonstrating clinical efficacy of B cell-modulating therapies was the use of rituximab in rheumatoid arthritis in 2004 (ref. 100). These early observations were subsequently confirmed in phase III studies that led to FDA approval for rituximab in patients with rheumatoid arthritis refractory to anti-TNF therapies¹⁰¹. During the following 12 years, rituximab was demonstrated to be efficacious and was approved by the FDA to treat patients with granulomatosis with polyangiitis (2011), microscopic polyangiitis (2011) and pemphigus vulgaris (2018)^{102,103}. The efficacy of rituximab in these B cell-mediated diseases revealed multiple non-mutually exclusive mechanisms by which B cells contribute to disease pathogenesis, including autoantibody-mediated cytotoxicity, immune complex-mediated end organ damage, amplification of immune activation through cytokine secretion and co-stimulation and tertiary lymphoid structure formation¹⁰⁴.

Conversely, rituximab was not efficacious in phase III clinical trials of systemic lupus erythematosus (SLE), lupus nephritis and primary progressive multiple sclerosis (PPMS)^{105,106}. Retrospective analysis of the phase III LUNAR study of rituximab in lupus nephritis demonstrated substantial variability in peripheral B cell depletion and that achievement of complete and prolonged depletion of CD19⁺ B cells was associated with complete clinical responses¹⁰⁷. Of note,

peripheral B cell depletion is an insensitive measurement of lymphoid or tissue B cell depletion¹⁰⁸. Furthermore, in MRL/lpr and NZB/W mouse models of autoimmune SLE, B cells are similarly refractory to depletion by anti-CD20 antibodies¹⁰⁹. For CD20 targeting, obinutuzumab was more effective than rituximab in depleting B cells and decreasing glomerulonephritis in MRL/lpr mice¹¹⁰. In the REGENCY phase III study of patients with lupus nephritis, obinutuzumab when added to mycophenolate and corticosteroids provided greater preservation of renal function when compared with patients receiving mycophenolate and corticosteroids alone¹¹¹.

In multiple sclerosis, rituximab provided substantial reduction in gadolinium-enhancing lesions and clinical relapses in a phase II clinical trial of patients with relapsing–remitting multiple sclerosis (RRMS)¹¹². However, rituximab was not efficacious in a phase III clinical trial of PPMS¹¹³. Anti-drug antibodies (ADAs) were detected in 26–37% of patients with multiple sclerosis following treatment with rituximab, although the presence of ADAs was not associated with any clinical adverse events¹¹⁴. This immunogenicity of rituximab likely reflects that it is a chimeric antibody with mouse variable domains and that patients with RRMS or PPMS do not receive concomitant immunosuppression, unlike patients with rheumatoid arthritis. Although safe for almost all patients treated, there are rare cases of rituximab-induced serum sickness in multiple sclerosis¹¹⁵. However, there are no predictive markers to identify before therapy patients who may develop these rare adverse events. In SLE, high rates of ADAs to rituximab have also been reported and are associated with earlier drug elimination¹¹⁶. Ocrelizumab and ofatumumab are humanized and human anti-CD20 antibodies, respectively, that have been approved by the FDA for treatment of RRMS^{117,118}. Ocrelizumab is also approved by the FDA for treatment of PPMS¹¹⁹. Thus, humanized and human antibodies can decrease the potential for ADAs, their potential negative impact on drug clearance and development of rare type III hypersensitivity reactions.

Bispecific antibodies to CD19 and CD20 are being tested in SLE and lupus nephritis (NCT05835986 and NCT05155345) and are anticipated to provide a better understanding of whether these more potent B cell-modulating therapies can deliver greater clinical efficacy. An exciting recent development is the emergence of CAR T cell therapy for treatment of autoimmune disorders, including SLE, lupus nephritis, inflammatory myositis and multiple sclerosis^{120,121}. We await larger clinical trials of these new scFv-directed cell therapies against CD19, BCMA and other B cell lineage markers.

HER2 targeting for breast cancer

At least 11 different antibody therapeutics have been approved for use in oncology for various tumours overexpressing the HER2 proto-oncogene, including breast, gastric, biliary tract, gastro-oesophageal cancers and non-small-cell lung cancer (NSCLC) (Fig. 4). Here, we focus on how advances in antibody engineering have improved clinical outcomes in the treatment of HER2⁺ breast cancer. The overexpression of HER2 as an oncogenic driver in ~20% of breast cancers provided a rationale for the development of antibody therapeutics, the first of which was trastuzumab¹²².

The humanized IgG1 antibody trastuzumab binds to subdomain IV of the HER2 extracellular domain (ECD), inhibits tumour cell proliferation and supports ADCC¹²³. Additionally, trastuzumab shows additivity or synergy with cytotoxic chemotherapy¹²⁴. Trastuzumab in combination with chemotherapy improved both progression-free and overall survival in HER2⁺ metastatic breast cancer¹²⁵. In 1998 trastuzumab was approved for the treatment of HER2⁺ metastatic breast

cancer. Trastuzumab was accompanied by the first ever FDA-approved companion diagnostic test (HercepTest) for detection of HER2 protein expression by immunohistochemistry¹²⁶. Subsequent clinical trials led to the approval of trastuzumab for the treatment of early HER2⁺ breast cancer¹²⁷.

The humanized IgG1 antibody pertuzumab binds to subdomain II of the HER2 ECD and prevents heterodimerization of HER2 with other HER family members, namely, EGFR, HER3 and HER4. The combination of trastuzumab and pertuzumab results in enhanced inhibition of downstream signalling¹²⁸. Clinical trials with trastuzumab plus pertuzumab administered by i.v. infusion in combination with chemotherapy, demonstrated greater clinical benefit than trastuzumab with chemotherapy across the spectrum of metastatic, adjuvant and neoadjuvant HER2⁺ breast cancers^{129–131}. A further advance was coformulation of trastuzumab with pertuzumab plus hyaluronidase-zzxf, thereby enabling s.c. delivery for treatment of HER2⁺ breast cancer in conjunction with chemotherapy¹³².

Polymorphisms in Fc receptors, notably FcγRIIIa, have been associated with improved trastuzumab treatment outcomes in some but not all studies^{133–135}. The goal of enhancing ADCC function prompted the development of margetuximab-cmkb (previously known as MGAH22 (ref. 136)). Margetuximab-cmkb is a chimeric IgG1 version of the mouse anti-HER2 monoclonal antibody, 4D5 (ref. 137), which is also the parent antibody of trastuzumab¹²³. Margetuximab-cmkb has five amino acid substitutions in its Fc region (F243L:R292P:Y300L:V305I:P396L) to enhance its affinity for the activation receptor, FcγRIIIa, and to decrease its affinity for the inhibitory receptor, FcγRIIb¹³⁶. The combination of margetuximab-cmkb plus chemotherapy in the SOPHIA study demonstrated a statistically significant improvement in progression-free

Table 1 | Enhancing the potential of antibodies for subcutaneous delivery

Strategy	Specific approaches
Increase s.c. dosing volume; current limit is ~2 ml	Inject subcutaneously at two or more sites per treatment, if needed, as exemplified by omalizumab, dupilumab, ixekizumab and belimumab Coformulate antibody with hyaluronidase to enable s.c. delivery of larger volumes (5–15 ml) as demonstrated for multiple approved antibodies: rituximab, trastuzumab, pertuzumab plus trastuzumab, daratumumab, atezolizumab and nivolumab
Increase s.c. dosing concentration; current limit is ~200 mg ml ⁻¹	Add excipients to reduce the viscosity, enhance the stability or solubility ^{168,233} Screen antibodies for favourable high-concentration properties: high solubility, low viscosity, low propensity for aggregation, no opalescence or gelation Engineer antibody variable domains ^{174–176} and/or Fc region ¹⁷⁷ to improve high-concentration properties, including reduced viscosity
Reduce required antibody dose	Engineer or screen antibodies for enhanced properties: Fc mutations such as M252Y:S254T:T256E ²³⁴ or M428L:N434S ¹⁶⁵ to increase the plasma half-life ^{235,236} and target exposure Sweeping ^{178,179} or recycling ^{180,181} antibodies to enable super-stoichiometric neutralization of the target antigens (see Fig. 5) Greater potency without compromising safety Increased bioavailability; requires improved experimental and computational methods for preclinical screening ²³⁷

s.c., subcutaneous.

survival compared with trastuzumab plus chemotherapy in patients with HER2⁺ advanced breast cancer previously treated with two or more anti-HER2 therapies¹³⁸. However, margetuximab-cmkb did not improve overall survival in the SOPHIA study.

Before trastuzumab, cytotoxic chemotherapies and surgery were the mainstay of treatment of HER2⁺ breast cancer. Whereas surgery remains a mainstay for therapy in early disease, chemotherapeutic regimens have a narrow therapeutic index that limits their potential efficacy. The first ADC introduced for the treatment of HER2⁺ breast cancer was ado-trastuzumab emtansine (T-DM1). This ADC consists of trastuzumab (T) covalently connected via a non-cleavable linker to a maytansinoid, DM1, that binds to microtubules, leading to mitotic disruption and apoptosis in HER2-overexpressing cell lines¹³⁹. After binding of cell surface HER2, T-DM1 is internalized through the endolysosomal pathway and undergoes degradation, releasing its cytotoxic payload to kill the tumour cell. T-DM1 prolonged both progression-free and overall survival and was less toxic and better tolerated by patients when compared with lapatinib and capecitabine in patients with HER2⁺ breast cancer previously treated with trastuzumab and taxanes¹⁴⁰. T-DM1 is also approved to treat recurrent disease after neoadjuvant therapy¹⁴¹.

ADC linker design and DAR impact the amount of payload delivered per tumour, as well as in vivo ADC stability, pharmacokinetics, efficacy and toxicity. Additionally, cell permeability of the released payload is likely necessary for efficient killing of bystander cells. T-DM1 has a mean DAR of ~3.5. Fam-trastuzumab deruxtecan-nxki (T-DXd) is a second-generation anti-HER2 ADC with a higher mean DAR (~8). In T-DXd, trastuzumab is conjugated through a cleavable and self-immolative amino methylene spacer to deruxtecan (Dxd)¹⁴². Dxd inhibits DNA topoisomerase I and thereby DNA replication resulting in cell death. T-DXd (previously known as DS-8201a) releases cell-permeable metabolites and exhibits bystander cell killing activity¹⁴³, whereas T-DM1 releases cell-impermeable metabolites and does not¹³⁹. T-DXd was initially approved for treatment of unresectable or metastatic HER2⁺ breast cancer^{144,145}. Surprisingly and importantly, T-DXd is also efficacious in metastatic breast cancers that express low levels of HER2 (ref. 146). Efficacy in this latter population may reflect a higher dose (5.4 mg kg⁻¹ every 3 weeks versus 3.6 mg kg⁻¹ every 3 weeks for T-DM1), higher DAR and potential bystander effect to overcome HER2 expression tumour heterogeneity. Subsequent approvals for T-DXd include HER2-low and HER2-ultralow unresectable or metastatic breast cancer, as well as HER2⁺ metastatic or recurrent breast cancer in the neoadjuvant or adjuvant setting¹⁴⁷. T-DXd is also approved for other tumour types, namely, NSCLC with an activating HER2 mutation and trastuzumab-experienced HER2⁺ gastric or gastro-oesophageal cancers¹⁴⁷. Another HER2-directed ADC, distamab vedotin, with a monomethyl auristatin E payload, was recently approved in China for treatment of HER2⁺ gastric cancer (see [Antibody Society](#) data).

Multiple additional HER2-targeted ADCs with diverse payloads and linker types are presently in development that will provide additional insights into how mechanistic differences may improve efficacy and minimize toxicities¹²². In addition, further understanding of the impact of ADCs on direct and indirect tumour cytotoxicity, the spatial distribution of HER2-high versus HER2-low targeted tumours and the potential impact of the stromal and immune compartments will also likely be important not only for next-generation drug design, but also for combination therapies that can modulate innate and acquired resistance mechanisms. Multiple investigational therapeutics are being

explored to test these concepts. The bispecific antibody IBI315 (targeting HER2 and PD1) enhances antitumour activity in preclinical studies of HER2⁺ gastric cancer and reached clinical trials for this indication (NCT04162327 and NCT05608785). Cinrebafusp alfa (PRS-343), is a bispecific fusion protein that combines a 4-1BB-targeting Anticalin and a HER2-targeting antibody. Cinrebafusp alfa demonstrated some antitumour activity in a phase I study and is currently in a phase II trial (NCT05190445) in HER2-low gastric and gastro-oesophageal junction adenocarcinoma. MT-5111 is an engineered toxin body: scFv targeting HER2 genetically fused to the enzymatically active de-immunized, ribosome-inactivating cytotoxic payload Shiga-like toxin A subunit. MT-5111 reached phase I clinical trial in HER2⁺ solid tumours (NCT04029922).

One biparatopic antibody (zanidatamab-hrii) and one bispecific antibody (zenocutuzumab-zbco, targeting HER2 and HER3) have been approved to target HER2⁺ cancers (Fig. 4). Zanidatamab-hrii binds to two distinct epitopes on subdomains II and IV of the HER2 ECD, thereby inhibiting HER2 homodimerization and heterodimerization with EGFR, HER3 or HER4 and downstream intracellular signalling¹⁴⁸. Additionally, zanidatamab-hrii facilitates HER2 internalization and subsequent downregulation, leading to a reduction of HER2 receptor on the surface of tumour cells. Zanidatamab-hrii also induces tumour cell killing via CDC, ADCC and ADCP. These mechanisms of action result in tumour growth inhibition and cell death in vitro and in vivo. Zanidatamab-hrii demonstrated clinical efficacy in multiple HER2⁺ tumour types, including biliary tract cancer, and is currently approved for this indication^{149,150}.

Chimeric proteins resulting from neuregulin 1 (NRG1) genetic fusions are rare (~0.2% prevalence) oncogenic drivers in some solid tumours, including NSCLC and pancreatic adenocarcinoma¹⁵¹. NRG1 fusions facilitate heterodimerization of HER3 with HER2, triggering cellular proliferation and oncogenic activity¹⁵². Zenocutuzumab-zbco is a common light chain⁵² bispecific antibody (to HER2 and HER3) that blocks NRG1 binding to HER3 and prevents heterodimerization with HER2. Zenocutuzumab demonstrated efficacy in a registrational phase II trial (eNRGy) in patients with advanced NRG1 fusion-positive cancer such as NSCLC and pancreatic cancer¹⁵³, leading to approval for these indications.

Just as discussed above for haematological cancers, TCEs and CAR T cells are also being explored for HER2⁺ malignancies. Indeed, there are 20 active clinical trials for HER2 CAR T cells as of May 2025 (<https://www.clinicaltrials.gov>). Unlike CD19, CD20 and BCMA, which are expressed almost exclusively on immune cells, HER2 is also expressed at lower levels on normal epithelial cells¹⁵⁴, as well as on cardiomyocytes¹⁵⁵. Treatment with trastuzumab is associated with a risk of cardiotoxicity, which is greatest in patients receiving concomitant anthracyclines¹⁵⁶. As technological advances increase the potency of antibody therapeutics, one must be watchful for adverse events associated with HER2 expression on normal tissue. For example, the death of one patient in a clinical trial with a HER2 CAR T cell apparently resulting from CRS has been reported¹⁵⁷. Avidity engineering provides a way to enhance the selectivity of TCEs, and potentially CAR T cells too, for efficient killing of tumour cells expressing high levels of cognate antigen while sparing normal cells expressing lower levels of antigen¹⁵⁸. For example, a HER2-targeting TCE was designed with 2+1 bispecific IgG format using two low-affinity HER2-binding arms. This avidity-engineered TCE efficiently killed tumour cells expressing high levels of HER2 with attenuated killing of cells expressing low levels of HER2. Such avidity engineering has the potential to reduce the adverse events from

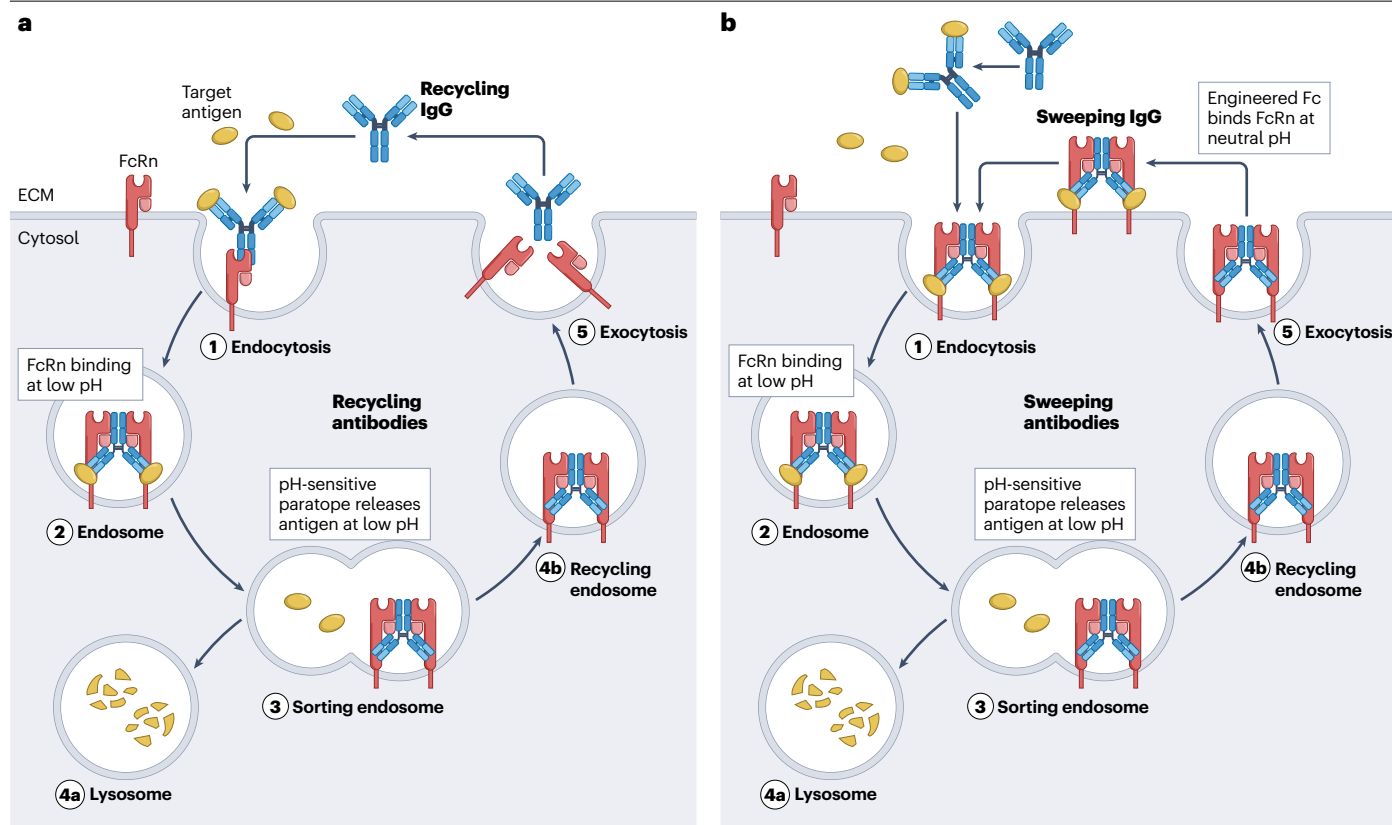


Fig. 5 | Recycling and sweeping antibodies facilitate super-stoichiometric clearance of their cognate target antigen. a, Recycling antibodies have an engineered paratope to bind to their cognate antigens in a pH-dependent manner. Under physiological conditions (pH 7.4), antibody–antigen interactions are strong. As the antibody is internalized via passive endocytosis, the drop in pH of the endosome results in Fc-mediated binding of neonatal Fc receptor (FcRn) while the engineered paratope releases the target antigen. The antibody

is then recycled back to the extracellular space and released, whereas the antigen is digested in the lysosome. **b,** Sweeping antibodies also have an engineered paratope to bind to target antigen in a pH-dependent manner. However, they contain additional Fc modifications that enhance binding to the FcRn at pH 7.4. These modifications allow antibodies to maintain a cycle of target antigen binding and deposition to the endosome for degradation while maintaining contacts with the FcRn at the surface of the cell, as well as in the endosome.

TCEs and increase the therapeutic index. Runimotamab, a HER2-targeting TCE that is engineered for avidity, is currently in a phase Ia/Ib clinical trial (NCT03448042) for HER2-expressing cancers. Further advances in logic-gated antibodies¹⁵⁹ that can distinguish between normal and cancer cells may further increase the therapeutic index of next-generation therapies.

Current practices and future trends for antibody delivery

As noted previously, most antibodies are administered by the i.v. or s.c. route (Fig. 2d). Below we discuss the strengths and limitations of i.v. and s.c. delivery, alongside opportunities to mitigate these limitations.

Strengths and limitations of intravenous antibody delivery

An attractive feature of i.v. delivery is its broad applicability for antibodies for many different clinical indications. Additionally, the plasma bioavailability following i.v. delivery is 100% by definition. However, a drawback of i.v. infusion is that treatment commonly requires at least 1 h and often multiple hours in an outpatient infusion centre or hospital, adding to the cost and inconvenience for patients¹⁶⁰. These multiple burdens associated with the i.v. delivery of antibodies

can decrease compliance and reduce patient access to treatment as observed during the COVID-19 pandemic¹⁶¹.

Antibody engineering can partially mitigate the burdens of i.v. delivery of antibodies by enabling longer intervals between doses. This concept is illustrated by the iterative design of antibodies targeting the complement component, C5, for chronic treatment of the haematological disorder paroxysmal nocturnal haemoglobinuria. The humanized anti-C5 antibody eculizumab¹⁶² has a terminal plasma half-life of 11–17 days, requiring frequent i.v. administration. A representative treatment regimen for eculizumab is weekly dosing for 5 weeks followed by maintenance therapy every 2 weeks¹⁶³. Ravulizumab-cwvz (previously known as ALXN-1210)¹⁶⁴ is an engineered version of eculizumab with two half-life-extending modifications. First, ravulizumab-cwvz has pH-dependent antigen binding that allows recycling (Fig. 5) within cells to mitigate target-mediated drug disposition. Second, ravulizumab-cwvz incorporates two Fc mutations (M428L:N434S)¹⁶⁵ to increase recycling efficiency by the immunoglobulin salvage receptor, neonatal Fc receptor (FcRn). Ravulizumab-cwvz has an extended terminal plasma half-life of 50–64 days in patients, allowing infrequent i.v. dosing, for example, a single loading dose followed by maintenance doses every 4–8 weeks¹⁶⁶.

Strengths and limitations of subcutaneous delivery

S.c. injection has multiple advantages over i.v. infusion, including much faster dosing (typically less than 10 min) the possibility of home administration, greater patient convenience and compliance, and lower health-care costs^{160,167}. Physical limitations of s.c. injection of antibodies typically include small dosing volumes (0.5–2 ml per injection site) and required high dosing concentrations (100–200 mg ml⁻¹)¹⁶⁸. Achieving high antibody concentration is often challenging owing to antibody self-interaction, which can result in high viscosity, insufficient solubility, aggregation, gelation and opalescence¹⁶⁹. An additional limitation of s.c. delivery is that the bioavailability of approved antibodies varies more than twofold (49–96%)¹⁰. Low bioavailability may undermine the feasibility of achieving the target dose required for s.c. delivery, as well as increasing the cost of goods. Moreover, the bioavailability of antibodies in humans is not yet routinely predictable preclinically using either computational or experimental methods, prompting an open challenge to develop better approaches¹⁷⁰.

Immunogenicity, including the elicitation of ADAs, is a potential risk for all protein therapeutics, including antibodies^{12,171}. ADAs on rare occasions are clinically significant in that they adversely affect the pharmacokinetics, pharmacodynamics, safety or efficacy of protein therapeutics^{12,172}. Systematic comparison of the immunogenicity of protein therapeutics ($n = 20$) revealed that the incidence of ADAs in most cases (70%) was similar following i.v. or s.c. delivery. However, the incidence of ADAs was slightly higher in a few examples for s.c. injection (20% of cases) and substantially higher in rare instances (10% of cases)¹⁶⁰. Given these small differences in observed ADA rates, immunogenicity is not commonly a major consideration in deciding between i.v. and s.c. administration for antibody and other protein therapeutics.

The plasma pharmacokinetics of antibodies are inevitably different following i.v. or s.c. administration¹⁶⁰. For example, the maximal plasma concentration ($C_{\max}$) will be lower after s.c. injection. Furthermore, $t_{\max}$ (the time to reach $C_{\max}$) is much longer, taking multiple days compared with hours¹⁷³. These pharmacokinetic differences for s.c. delivery can be either an advantage or disadvantage depending upon the circumstances¹⁶⁰. On the positive side, a lower $C_{\max}$ after s.c. injection may lower the risk of CRS for TCEs¹⁶⁰. A downside is the difficulty in monitoring and managing infusion-related reactions that are often more common and severe for the first dose, especially if these reactions occur days after a s.c. injection. This safety concern can be mitigated by i.v. infusion for the first dose followed by s.c. injection for subsequent doses, as exemplified by approved dosing regimens for several different antibodies, including caplacizumab-yhdp, mirikizumab-mrkz, pozelimab-bbfg and crovalimab-akkz.

Multiple complementary antibody engineering and formulation strategies are possible to enhance the potential of antibodies for s.c. delivery (Table 1). For example, coformulation of antibodies with hyaluronidase to hydrolyse hyaluronic acid in connective tissue enables s.c. injection volumes to be increased from 2 ml to at least 15 ml, as demonstrated for at least six approved antibody therapeutics (Table 1). In 1998, trastuzumab, delivered by i.v. infusion, was approved for HER2⁺ metastatic breast cancer. In 2013, trastuzumab coformulated with hyaluronidase-oysk and delivered by s.c. injection was approved for HER2⁺ early breast cancer (Fig. 4). Furthermore, antibody engineering can be used to improve antibody high-concentration properties such as reducing viscosity^{174–177}, thereby facilitating s.c. delivery. Additionally, engineering can be used to increase the pharmacodynamic effect of antibodies and thereby lower the required antibody dose (Table 1). For example, engineering antibodies with sweeping^{178,179} or recycling^{180,181}

properties (Fig. 5) enables super-stoichiometric neutralization of the target antigen by the antibody. The anti-C5 antibody, crovalimab-akkz (previously known as SKY-59), was engineered for recycling^{181,182} and is approved for the treatment of paroxysmal nocturnal haemoglobinuria by i.v. infusion for the first dose followed by s.c. injection for subsequent doses.

Future opportunities for antibody therapeutics

This Review focuses on the 50-year odyssey from the advent of hybridoma technology in 1975 (ref. 1) to the present era of antibody therapeutics (Fig. 2) and their clinical application to targeting B cells (Fig. 3) and HER2⁺ cancers (Fig. 4). Emerging trends for antibody therapeutics with broad potential impact have been extensively reviewed elsewhere^{10,13}, including for bispecifics^{22,23} and ADCs²¹. Here, we opine briefly on four major areas of innovation that hold much promise for future antibody therapeutics: first, applications of artificial intelligence to antibody design; second, conditional antigen-binding activity to enhance the therapeutic index of anticancer antibodies; third, exploration of immunoglobulin isotypes beyond IgG; and fourth, more facile generation of agonist antibodies. Finally, we comment on a few grand challenges with antibodies for which even partial solutions may have profound benefit for next-generation antibody therapeutics: efficient delivery of antibodies into the central nervous system; engineering antibodies for oral administration; and engineering antibodies with low risk for immunogenicity in patients.

Applications of artificial intelligence to antibody design

The 2024 Nobel Prize in Chemistry¹⁸³ was awarded to David Baker for “computational protein design” jointly with John Jumper and Demis Hassabis for “protein structure prediction” with AlphaFold¹⁸⁴. This Nobel Prize recognizes the enormous potential of artificial intelligence tools for protein sciences, including the design of antibody and other protein therapeutics. Indeed, computational design can endow proteins with novel functionality that can be difficult to accomplish using rational engineering or directed evolution. This is highlighted by the de novo design of high-affinity (picomolar range) protein binders reported by David Baker and collaborators¹⁸⁵, as well as the computational design of zinc finger domains to customize DNA-binding motifs¹⁸⁶.

However, there are many steps beyond the design of novel antibodies and other proteins to ensure that they have drug-like properties. Indeed, the design of antibody therapeutics is quintessentially one of optimization across multiple desired attributes. First, one needs the desired binding specificity, affinity, biological activities plus pharmacokinetics. Additionally, favourable developability characteristics are desirable, including chemical and thermal stability. Desirable attributes for s.c. delivery include favourable high-concentration properties, high bioavailability and long-term (multi-month) stability at ambient temperatures. Low risk for immunogenicity is also a valuable attribute for engineered protein and antibody therapeutics¹². Antibody engineers can finely tune these multiple properties to produce molecules with drug-like properties with great success, albeit using resource-intensive methods¹⁰.

The computational design and optimization of antibodies for affinity^{187–189}, developability^{190,191}, including low viscosity¹⁹², humanization¹⁹³ and immunogenicity¹⁹⁴, is already under way, and the de novo design of VHH antibodies has also been reported¹⁹⁵. The computational methods that underlie these breakthroughs continue to evolve, as reviewed elsewhere¹⁹⁶. However, the ultimate promise of computational antibody design is integration of all of the

aforementioned properties into an antibody from conception – in silico multi-parameter optimization. Artificial intelligence methods have the potential to generate or optimize antibodies in ways that are difficult or not possible with extant experimental approaches, as well as reducing the time and/or cost to generate clinical candidates¹⁹⁷.

Engineering antibodies with conditional antigen binding

Antibody therapeutics have grown in their impact in oncology over nearly three decades as illustrated in this Review with multiple generations of antibodies targeting B cells and HER2. Much current research is focused on increasing efficacy and reducing toxicity for anticancer antibodies¹¹, including engineering antibodies for conditional antigen binding. For example, antibody prodrugs (Probody) have been generated by fusion of antibody variable domains with peptides to attenuate antigen binding¹⁹⁸. Selective activation of such antibody prodrugs by tumour-associated proteases has the potential to enhance the selectivity of antibodies for tumours over normal tissue and thereby increase the therapeutic index¹¹. Alternative strategies for engineering antibodies for enhanced binding in tumours include exploiting the slightly acidic pH in the tumour microenvironment^{180,199}, or binding to extracellular ATP^{200,201} or calcium ions²⁰². Clinical testing of the potential benefits of conditional antigen binding by antibodies is in progress (for example, [NCT06265688](#) and [IRAS ID 270978](#)).

Immunoglobulin isotypes

A large proportion of approved antibodies have human IgG1 (57%), IgG2 (7%) or IgG4 (22%) (see [Antibody Society](#) data) heavy chain constant domains. The Fc region of IgG therapeutics is commonly engineered to modify one or more functions, as comprehensively reviewed by Wilkinson and Hale⁹. The common selection of the human IgG1 isotype for antibody therapeutics likely reflects in part the ease with which IgG1 Fc functions have been successfully tailored for specific clinical applications, including enhancement or attenuation of effector functions, extension or reduction of plasma half-life, heterodimerization to make bispecific antibodies, ablation of protein A binding and enablement of site-specific conjugation for ADCs (see [Antibody Society](#) data)⁹. There are no approved IgG3 therapeutics, suggesting that the advantages of this isotype are outweighed by potential limitations. Proposed advantages for IgG3 include high affinity for activating Fcγ receptors, effective complement fixation and a long hinge that may improve suitability for low-abundance targets²⁰³. Potential downsides for IgG3 include rapid degradation, short plasma half-life and immunogenicity risk stemming from multiple polymorphisms (allotypes)²⁰³.

Recent years have seen growing interest in evaluation of other human immunoglobulin isotypes in the clinic, including IgM^{204,205}, IgA²⁰⁶ and IgE^{207,208}. These alternative immunoglobulin isotypes warrant exploration as they have distinctly different properties from IgG, including their avidity, effector function profile and biodistribution in human as illustrated here with IgM, which is reviewed in greater detail elsewhere^{204,205}. IgM forms pentamers or alternatively hexamers with 10 or 12 antigen-binding sites, respectively, compared with only two antigen-binding sites for IgG. The large number of binding sites for IgM can be advantageous over IgG in supporting high-avidity antigen binding, target clustering and very efficient complement activation. For example, reformatting an anti-SARS-CoV-2 spike protein IgG into an IgM pentamer increased the potency for neutralization of SARS-CoV-2 in vitro 230-fold (ref. 209). Major challenges to development of IgM as therapeutics, include their large size (900–950 kDa for pentamers and 1,050–1,150 kDa for hexamers), lack of protein A binding, limited

stability and short pharmacokinetic half-life in human. Although a human IgM isotype antibody, nebacumab (anti-endotoxin) (Fig. 2c) was approved in several European countries for treatment of sepsis (1991), it was quickly withdrawn from the market because of lack of efficacy and increased mortality²¹⁰.

An alternative strategy to the direct development of IgM, IgA and IgE as therapeutics is to engineer IgG to endow it with properties from other immunoglobulin isotypes. For example, the HexaBody platform for hexamerization of IgG was designed to mimic IgM²¹¹. A HexaBody targeting CD38 (GEN3014) is in a phase I/II trial for R/R haematological malignancies ([NCT04824794](#)). A HexaBody targeting OX40 (GEN1055) is in a phase I/II trial for malignant solid tumours as a monotherapy and in combination with pembrolizumab ([NCT06391775](#)). Another approach to expanding the functions of IgG is to design hybrids with other isotypes. For example, a hybrid IgE–IgG1 (IgEG) of trastuzumab was recently engineered to combine the functions of IgG1 and IgE²¹². In vitro trastuzumab IgEG showed a broader range of immune functions than either corresponding IgG1 or IgE. In mice the pharmacokinetic half-life of trastuzumab IgEG (3.4 days) was intermediate between that of the corresponding IgG1 (14.1 days) and that of IgE (0.008 days), whereas the antitumour activity of the IgEG was comparable to that of both IgE and IgG1 versions²¹².

Agonist antibodies

Antibody therapeutics binding to either receptors or their cognate ligands are well established as antagonists of receptor–ligand interactions²¹³. In contrast, the development of agonist antibodies for activation of receptors^{214,215}, including cytokine mimetic antibodies²¹⁶, is at a much earlier stage, reflecting that they can be challenging to identify. Cytokine mimetic antibodies have both potential advantages and limitations as therapeutics compared with engineered cytokines²¹⁷. Advantages of cytokine mimetic antibodies include typically more favourable developability properties, potential for new biological properties, avoidance of natural inhibitors and readily tunable pharmacokinetic properties. Additionally, any ADAs elicited by the cytokine mimetic antibodies are unlikely to neutralize endogenous cytokines. Disadvantages of cytokine mimetic antibodies are that they can be more difficult to design or discover and they may not recapitulate all of the desired natural biological properties of the cytokine.

Delivery of antibodies to the central nervous system

Beyond enhancing antibodies for s.c. administration, delivery to organs and tissues, including the central nervous system²¹⁸, is a major research focus with antibodies with substantial therapeutic potential¹⁰. For example, engineering antibodies with bispecificity, such as binding to a target antigen and to the transferrin receptor (TfR), can enhance uptake across the blood–brain barrier into the brain several fold as demonstrated in non-human primates²¹⁹. Trontinemab, a 2+1 bispecific antibody (targeting amyloid-β and TfR)²²⁰ that is also known as a Brainshuttle, reduced amyloid plaques in the brain in a phase Ib/IIa clinical trial in Alzheimer disease ([NCT04639050](#)) with phase III trials now being planned. Potential future applications of bispecific antibodies with enhanced delivery to the brain include the treatment of brain cancers or other cancers that have metastasized to the brain.

Clinical proof of concept for the use of TfR to enhance uptake into the central nervous system has been achieved in the context of enzyme replacement therapy. Mucopolysaccharidosis type II (Hunter syndrome) is a rare X-linked recessive lysosomal storage disease caused by a deficiency of iduronate 2-sulfatase that leads to a build-up of

Glossary

Adverse events

Unfavourable medical occurrences in patients, irrespective of whether they are causally related to the treatment.

Antigen

Any molecule that induces a specific immune response in animals or humans, such as the elicitation of cognate antibodies.

Bioavailability

Fraction (percentage) of the administered dose of a therapeutic that reaches systemic circulation.

Biparatopic

Describes a type of bispecific antibody that binds two non-overlapping locations (epitopes) on the same antigen using two unique paratopes.

Compliance

The extent to which a patient takes a therapeutic as prescribed by their physician.

Cost of goods

Direct cost of producing the therapeutic in its final formulated form.

Cytokine release syndrome

A severe systemic inflammatory response characterized by the excessive release of pro-inflammatory cytokines, leading to symptoms that include fever, hypotension and multi-organ dysfunction.

glycosaminoglycans in cells, resulting in various symptoms and complications. Pabinafusp alfa is a blood–brain barrier penetrant fusion protein consisting of an antibody that binds to TfR fused to iduronate 2-sulfatase²²¹. Systemic administration of pabinafusp alfa utilizes TfR-mediated transcytosis to increase iduronate 2-sulfatase levels in the central nervous system. In 2021, pabinafusp alfa was approved in Japan to treat mucopolysaccharidosis type II. This approach is likely to be used for other enzyme replacement therapies that require central nervous system penetration.

Oral delivery of antibodies

The oral delivery of antibodies and other proteins is a long-sought-after pharmaceutical goal to increase patient compliance and convenience

Developability

Desirable drug-like properties of antibody therapeutics, including feasibility of manufacture, stability during storage, ease of administration and favourable pharmacological properties in patients, and excluding target binding.

Effector functions

Fc-mediated antibody properties that allow target cell depletion via immune mechanisms such as antibody-dependent cell-mediated cytotoxicity, antibody-dependent cell-mediated phagocytosis and complement-dependent cytotoxicity.

Epitopes

Specific regions of an antigen that contact a cognate antibody.

Paratope

Specific location on the surface of antibody variable domain that contacts the cognate antigen.

Target exposure

The level of a therapeutic that reaches and interacts with its intended biological target in the dosed animal or human.

Target-mediated drug disposition

Target binding by a therapeutic that leads to its internalization and degradation.

Therapeutic index

The ratio between the dose that produces a therapeutic effect and the dose that produces toxic effects, indicating the safety and efficacy of the treatment.

by circumventing the discomfort associated with parenteral delivery methods and by minimizing related toxicities^{222,223}. However, oral delivery of proteins is fraught with substantial obstacles, including inactivation of proteins by low pH and proteases, as well as inefficient antibody release at the desired site in the gastrointestinal tract. Further and compounding challenges include therapeutic doses limited by pill burden and substantial permeability barriers to uptake from the gastrointestinal tract into systemic circulation, predominantly involving mucus and epithelia. Steps towards enabling the oral delivery of proteins include the use of nanoparticles, permeation enhancers, hydrogels, microneedles, microemulsion, protease inhibitors and cell-penetrating peptides, as reviewed extensively elsewhere^{222,223}. For example, sophisticated and ingestible medical devices have been used preclinically for systemic delivery of antibodies, peptides and small molecules (adalimumab, GLP1 analogue, insulin and adrenaline) by auto-injection through the gut mucosa^{224,225}. At least 13 different protein and peptide therapeutics, albeit no antibodies yet, are approved for oral delivery¹⁰, including semaglutide, cyclosporine, desmopressin acetate and octreotide²²³. Recently a VHH antibody targeting IL-23 receptor was engineered for high affinity and protease stability for oral delivery²²⁶. This anti-IL-23 receptor VHH was comparably efficacious in mouse models of colitis following oral gavage to a systemically administered anti-IL-23 receptor IgG. In cynomolgus monkeys, a single pill of encapsulated VHH gave sustained IL-23 receptor inhibition in the peripheral blood for at least 24 hours²²⁶. Nevertheless, further progress is needed to realize the full potential of oral delivery for transformative antibody and other protein therapeutics.

Engineering antibodies with low risk for immunogenicity

A major focus in the early years of antibody therapeutics was in overcoming the limitations of murine antibodies in patients, including the immunogenicity of these foreign proteins (Box 1). In recent years, immunogenicity risk assessment and mitigation have emerged as urgent requirements in the design and optimization of antibody and other protein therapeutics^{12,171,227}. A plethora of powerful protein engineering methods now exist to improve the existing properties of antibodies or to endow them with new functions unknown to nature¹⁰. Unfortunately, the potential for such extensive engineering of antibodies to elicit ADAs that compromise safety and/or efficacy is poorly understood. For some investigational antibodies, such as bococizumab²²⁸, a high incidence of ADAs was a major factor in the decision to terminate further clinical development. Immunogenicity mitigation strategies include engineering out T cell epitopes²²⁹ and improving biophysical properties as exemplified by bococizumab²³⁰.

In contrast to the highly immunogenic antibody bococizumab, the extensively engineered bispecific antibody emicizumab-kxwh (mentioned above) had a desirably low incidence of ADAs (5.1%) in phase III clinical trials²³¹. This low ADA rate may reflect in part the use of a common light chain⁵² in the design of emicizumab and/or T cell epitope identification and removal²³². Emicizumab demonstrates that minimization of the incidence of ADAs in the clinic can be achieved, even with an extensively engineered bispecific antibody. However, more routine and predictable immunogenicity risk assessment and mitigation for engineered antibodies and other protein therapeutics is still an urgent and much sought-after goal for drug developers¹².

Perspectives

Georges Köhler and César Milstein concluded their landmark hybridoma publication¹ with the prophesy that “Such cultures could be

valuable for medical and industrial use.” The vision of these pioneers has been robustly realized, thanks to the ingenuity and perseverance of several generations of basic, translational and clinical researchers plus biotechnologists, as well as intrepid patients and their families. Remarkable progress with antibody therapeutics over many decades (Fig. 2) bodes well for even greater benefit to patients in the future with an expanding array of antibody-based modalities.

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References

- Köhler, G. & Milstein, C. Continuous cultures of fused cells secreting antibody of predefined specificity. *Nature* **256**, 495–497 (1975).
Ground-breaking invention of hybridoma technology that was a key step towards this modern era of antibody-based therapeutics.
- Bradbury, A. & Plückthun, A. Reproducibility: standardize antibodies used in research. *Nature* **518**, 27–29 (2015).
- Uhr, J. W. The 1984 Nobel Prize in Medicine. *Science* **226**, 1025–1028 (1984).
- Boulianne, G. L., Hozumi, N. & Shulman, M. J. Production of functional chimaeric mouse/human antibody. *Nature* **312**, 643–646 (1984).
Together with ref. 5 is the first demonstration of chimerization of a murine monoclonal antibody.
- Morrison, S. L., Johnson, M. J., Herzenberg, L. A. & Oi, V. T. Chimeric human antibody molecules: mouse antigen-binding domains with human constant region domains. *Proc. Natl Acad. Sci. USA* **81**, 6851–6855 (1984).
Together with ref. 4 is the first demonstration of chimerization of a murine monoclonal antibody.
- Jones, P. T., Dear, P. H., Foote, J., Neuberger, M. S. & Winter, G. Replacing the complementarity-determining regions in a human antibody with those from a mouse. *Nature* **321**, 522–525 (1986).
First demonstration of humanization of a murine antibody that, together with refs. 7 and 8, eventually led to at least 105 approved humanized antibody therapeutics.
- Riechmann, L., Clark, M., Waldmann, H. & Winter, G. Reshaping human antibodies for therapy. *Nature* **332**, 323–327 (1988).
- Verhoeven, M., Milstein, C. & Winter, G. Reshaping human antibodies: grafting an antilysozyme activity. *Science* **239**, 1534–1536 (1988).
- Wilkinson, I. & Hale, G. Systematic analysis of the varied designs of 819 therapeutic antibodies and Fc fusion proteins assigned international nonproprietary names. *MAbs* **14**, 2123299 (2022).
- Carter, P. J. & Rajpal, A. Designing antibodies as therapeutics. *Cell* **185**, 2789–2805 (2022).
- Paul, S. et al. Cancer therapy with antibodies. *Nat. Rev. Cancer* **24**, 399–426 (2024).
- Carter, P. J. & Quarmby, V. Immunogenicity risk assessment and mitigation for engineered antibody and protein therapeutics. *Nat. Rev. Drug Discov.* **23**, 898–913 (2024).
- Qian, L. et al. The dawn of a new era: targeting the “undruggables” with antibody-based therapeutics. *Chem. Rev.* **123**, 7782–7853 (2023).
- Brown, D. G., Wobst, H. J., Kapoor, A., Kenna, L. A. & Southall, N. Clinical development times for innovative drugs. *Nat. Rev. Drug Discov.* **21**, 793–794 (2022).
- Genetta, T. B. & Mauro, V. F. ABCIXIMAB: a new antiaggregant used in angioplasty. *Ann. Pharmacother.* **30**, 251–257 (1996).
- Vincenti, F. Daclizumab: novel biologic immunoprophylaxis for prevention of acute rejection in renal transplantation. *Transplant. Proc.* **31**, 2206–2207 (1999).
- Bain, B. & Brazil, M. Adalimumab. *Nat. Rev. Drug Discov.* **2**, 693–694 (2003).
- US Food and Drug Administration. Prescribing information, HUMIRA® (adalimumab) injection, for subcutaneous use. https://www.accessdata.fda.gov/drugsatfda_docs/label/2023/125057s423lbl.pdf (2023).
- Gibbons, J. B., Laber, M. & Bennett, C. L. Humira: the first \$20 billion drug. *Am. J. Manag. Care* **29**, 78–80 (2023).
- Crescioli, S. et al. Antibodies to watch in 2025. *MAbs* **17**, 2443538 (2025).
Most recent of a regular series of articles reviewing recently approved and registration stage antibody therapeutics.
- Dumontet, C., Reichert, J. M., Senter, P. D., Lambert, J. M. & Beck, A. Antibody–drug conjugates come of age in oncology. *Nat. Rev. Drug Discov.* **22**, 641–661 (2023).
- Goebeler, M. E., Stuhler, G. & Bargou, R. Bispecific and multispecific antibodies in oncology: opportunities and challenges. *Nat. Rev. Clin. Oncol.* **21**, 539–560 (2024).
- Klein, C., Brinkmann, U., Reichert, J. M. & Kontermann, R. E. The present and future of bispecific antibodies for cancer therapy. *Nat. Rev. Drug Discov.* **23**, 301–319 (2024).
- Pirkalkhoran, S. et al. Bioengineering of antibody fragments: challenges and opportunities. *Bioengineering* **10**, 122 (2023).
- Silver, A. B., Leonard, E. K., Gould, J. R. & Spangler, J. B. Engineered antibody fusion proteins for targeted disease therapy. *Trends Pharmacol. Sci.* **42**, 1064–1081 (2021).
- Larbouret, C., Gros, L., Pèlerin, A. & Chardès, T. Improving biologics’ effectiveness in clinical oncology: from the combination of two monoclonal antibodies to oligoclonal antibody mixtures. *Cancers* **13**, 4620 (2021).
- Cappell, K. M. & Kochenderfer, J. N. Long-term outcomes following CAR T cell therapy: what we know so far. *Nat. Rev. Clin. Oncol.* **20**, 359–371 (2023).
- Moolten, F. L. & Cooperband, S. R. Selective destruction of target cells by diphtheria toxin conjugated to antibody directed against antigens on the cells. *Science* **169**, 68–70 (1970).
First demonstration of an immunotoxin for directed killing of target cells.
- Casi, G. & Neri, D. Antibody–drug conjugates and small molecule–drug conjugates: opportunities and challenges for the development of selective anticancer cytotoxic agents. *J. Med. Chem.* **58**, 8751–8761 (2015).
- Sedalacek, H.-H. et al. (eds) *Antibodies as Carriers of Cytotoxicity. Contributions to Oncology* Vol. 43 (eds Huber, P. & Queisser, W.) (Karger, 1992).
- Mahalingaiah, P. K. et al. Potential mechanisms of target-independent uptake and toxicity of antibody–drug conjugates. *Pharmacol. Ther.* **200**, 110–125 (2019).
- Doronina, S. O. et al. Enhanced activity of monomethylauristatin F through monoclonal antibody delivery: effects of linker technology on efficacy and toxicity. *Bioconjug. Chem.* **17**, 114–124 (2006).
- Doronina, S. O. et al. Development of potent monoclonal antibody auristatin conjugates for cancer therapy. *Nat. Biotechnol.* **21**, 778–784 (2003).
First publication of ADCs with auristatin payloads, which has led to the approval of at least nine such ADCs.
- Bross, P. F. et al. Approval summary: gemtuzumab ozogamicin in relapsed acute myeloid leukemia. *Clin. Cancer Res.* **7**, 1490–1496 (2001).
- Norsworthy, K. J. et al. FDA approval summary: mylotarg for treatment of patients with relapsed or refractory CD33-positive acute myeloid leukemia. *Oncologist* **23**, 1103–1108 (2018).
- Junutula, J. R. et al. Site-specific conjugation of a cytotoxic drug to an antibody improves the therapeutic index. *Nat. Biotechnol.* **26**, 925–932 (2008).
- Lehar, S. M. et al. Novel antibody–antibiotic conjugate eliminates intracellular *S. aureus*. *Nature* **527**, 323–328 (2015).
- Dragovich, P. S. Degradable antibody conjugates. *Chem. Soc. Rev.* **51**, 3886–3897 (2022).
- Poudel, Y. B., Thakore, R. R. & Chekler, E. P. The new frontier: merging molecular glue degrader and antibody–drug conjugate modalities to overcome strategic challenges. *J. Med. Chem.* **67**, 15996–16001 (2024).
- Nisonoff, A., Wissler, F. C. & Lipman, L. N. Properties of the major component of a peptic digest of rabbit antibody. *Science* **132**, 1770–1771 (1960).
First demonstration of the concept of bispecific antibodies.
- Seimetz, D., Lindhofer, H. & Bokemeyer, C. Development and approval of the trifunctional antibody catumaxomab (anti-EpCAM x anti-CD3) as a targeted cancer immunotherapy. *Cancer Treat. Rev.* **36**, 458–467 (2010).
First approved bispecific: catumaxomab (targeting EpCAM and CD3; TCE).
- Milstein, C. & Cuello, A. C. Hybrid hybridomas and their use in immunohistochemistry. *Nature* **305**, 537–540 (1983).
- Staerz, U. D., Kanagawa, O. & Bevan, M. J. Hybrid antibodies can target sites for attack by T cells. *Nature* **314**, 628–631 (1985).
First demonstration of the concept of using bispecific antibodies to direct the killing of tumour cells by T cells, leading to at least nine such approved TCEs.
- Underwood, D. J., Bettencourt, J. & Jawad, Z. The manufacturing considerations of bispecific antibodies. *Expert Opin. Biol. Ther.* **22**, 1043–1065 (2022).
- Brinkmann, U. & Kontermann, R. E. The making of bispecific antibodies. *MAbs* **9**, 182–212 (2017).
- Spies, C., Zhai, Q. & Carter, P. J. Alternative molecular formats and therapeutic applications for bispecific antibodies. *Mol. Immunol.* **67**, 95–106 (2015).
- Mack, M., Riethmüller, G. & Kufer, P. A small bispecific antibody construct expressed as a functional single-chain molecule with high tumor cell cytotoxicity. *Proc. Natl Acad. Sci. USA* **92**, 7021–7025 (1995).
First demonstration of a TCE bispecific antibody (CD19 and CD3) in tandem scFv (BiTE) format that eventually led to blinatumomab.
- Ridgway, J. B., Presta, L. G. & Carter, P. ‘Knobs-into-holes’ engineering of antibody C_H3 domains for heavy chain heterodimerization. *Protein Eng.* **9**, 617–621 (1996).
First demonstration of ‘knobs-into-holes’ technology that after refinement (ref. 49 and ref. 52), has been widely used for Fc heterodimerization, including for at least three approved bispecific antibodies: mosunetuzumab, glofitamab-gxhm and faricimab-svoa.
- Atwell, S., Ridgway, J. B., Wells, J. A. & Carter, P. Stable heterodimers from remodeling the domain interface of a homodimer using a phage display library. *J. Mol. Biol.* **270**, 26–35 (1997).
- Schaefer, W. et al. Immunoglobulin domain crossover as a generic approach for the production of bispecific IgG antibodies. *Proc. Natl Acad. Sci. USA* **108**, 11187–11192 (2011).
First demonstration of CrossMab technology that has been used to generate at least two approved bispecific antibodies: glofitamab-gxhm and faricimab-svoa.
- Surowka, M., Schaefer, W. & Klein, C. Ten years in the making: application of CrossMab technology for the development of therapeutic bispecific antibodies and antibody fusion proteins. *MAbs* **13**, 1967714 (2021).
- Merchant, A. M. et al. An efficient route to human bispecific IgG. *Nat. Biotechnol.* **16**, 677–681 (1998).
First use of common light chains, a technology subsequently used for at least four approved bispecific antibodies: emicizumab, odronextamab, zenocutuzumab-zbco and linvoseltamab.

53. Labrijn, A. F. et al. Efficient generation of stable bispecific IgG1 by controlled Fab-arm exchange. *Proc. Natl Acad. Sci. USA* **110**, 5145–5150 (2013).
First demonstration of DuoBody technology that has been used for at least four approved bispecific antibodies: amivantamab-vmjw, epcoritamab-bysp, talquetamab-tgvs and teclistamab-cqyv.
54. Von Kreudenstein, T. S. et al. Improving biophysical properties of a bispecific antibody scaffold to aid developability: quality by molecular design. *MAbs* **5**, 646–654 (2013).
55. Yao, Y., Hu, Y. & Wang, F. Trispecific antibodies for cancer immunotherapy. *Immunol. Rev.* **169**, 389–399 (2023).
56. Cheson, B. D. & Leonard, J. P. Monoclonal antibody therapy for B-cell non-Hodgkin's lymphoma. *N. Engl. J. Med.* **359**, 613–626 (2008).
57. Reff, M. E. et al. Depletion of B cells *in vivo* by a chimeric mouse human monoclonal antibody to CD20. *Blood* **83**, 435–445 (1994).
58. Alduaij, W. et al. Novel type II anti-CD20 monoclonal antibody (GA101) evokes homotypic adhesion and actin-dependent, lysosome-mediated cell death in B-cell malignancies. *Blood* **117**, 4519–4529 (2011).
59. Ghosh, A. et al. Decoding the molecular interplay of CD20 and therapeutic antibodies with fast volumetric nanoscopy. *Science* **387**, eadq4510 (2025).
60. Weng, W. K. & Levy, R. Two immunoglobulin G fragment C receptor polymorphisms independently predict response to rituximab in patients with follicular lymphoma. *J. Clin. Oncol.* **21**, 3940–3947 (2003).
61. Carttron, G. et al. Therapeutic activity of humanized anti-CD20 monoclonal antibody and polymorphism in IgG Fc receptor FcγRIIIa gene. *Blood* **99**, 754–758 (2002).
First demonstration of the clinical activity of an antibody (rituximab) correlating with polymorphisms in an Fcγ receptor (FcγRIIIA).
62. Mossner, E. et al. Increasing the efficacy of CD20 antibody therapy through the engineering of a new type II anti-CD20 antibody with enhanced direct and immune effector cell-mediated B-cell cytotoxicity. *Blood* **115**, 4393–4402 (2010).
63. Cheson, B. D. et al. Overall survival benefit in patients with rituximab-refractory indolent non-Hodgkin lymphoma who received obinutuzumab plus bendamustine induction and obinutuzumab maintenance in the GADOLIN study. *J. Clin. Oncol.* **36**, 2259–2266 (2018).
64. Goede, V. et al. Obinutuzumab plus chlorambucil in patients with CLL and coexisting conditions. *N. Engl. J. Med.* **370**, 1101–1110 (2014).
65. Townsend, W. et al. Obinutuzumab versus rituximab immunochemotherapy in previously untreated NHL: final results from the GALLIUM study. *Hemasphere* **7**, e919 (2023).
66. Vitolo, U. et al. Obinutuzumab or rituximab plus cyclophosphamide, doxorubicin, vincristine, and prednisone in previously untreated diffuse large B-cell lymphoma. *J. Clin. Oncol.* **35**, 3529–3537 (2017).
67. Davies, A. et al. Obinutuzumab in the treatment of B-cell malignancies: a comprehensive review. *Future Oncol.* **18**, 2943–2966 (2022).
68. Marcus, R. et al. Obinutuzumab for the first-line treatment of follicular lymphoma. *N. Engl. J. Med.* **377**, 1331–1344 (2017).
69. Lazar, G. A. et al. Engineered antibody Fc variants with enhanced effector function. *Proc. Natl Acad. Sci. USA* **103**, 4005–4010 (2006).
Extensive mutational analysis of IgG1 Fc to tailor effector functions.
70. Salles, G. et al. Tafasitamab plus lenalidomide in relapsed or refractory diffuse large B-cell lymphoma (L-MIND): a multicentre, prospective, single-arm, phase 2 study. *Lancet Oncol.* **21**, 978–988 (2020).
71. Duell, J. et al. Tafasitamab for patients with relapsed or refractory diffuse large B-cell lymphoma: final 5-year efficacy and safety findings in the phase II L-MIND study. *Haematologica* **109**, 553–566 (2024).
72. Hiraga, J. et al. Down-regulation of CD20 expression in B-cell lymphoma cells after treatment with rituximab-containing combination chemotherapies: its prevalence and clinical significance. *Blood* **113**, 4885–4893 (2009).
73. Rehman, R. U., Anjum, A. F. & Fatima, R. Tarlatamab and the future of immunotherapy: a new approach to small cell lung cancer. *Curr. Ther. Res. Clin. Exp.* **102**, 100773 (2025).
74. Kantarjian, H. et al. Blinatumomab versus chemotherapy for advanced acute lymphoblastic leukemia. *N. Engl. J. Med.* **376**, 836–847 (2017).
75. Löffler, A. et al. A recombinant bispecific single-chain antibody, CD19×CD3, induces rapid and high lymphoma-directed cytotoxicity by unstimulated T lymphocytes. *Blood* **95**, 2098–2103 (2000).
76. Mocquot, P., Mossazadeh, Y., Lapiere, L., Pineau, F. & Despas, F. The pharmacology of blinatumomab: state of the art on pharmacodynamics, pharmacokinetics, adverse drug reactions and evaluation in clinical trials. *J. Clin. Pharm. Ther.* **47**, 1337–1351 (2022).
77. Elmeliegy, M. et al. Dosing strategies and quantitative clinical pharmacology for bispecific T-cell engagers development in oncology. *Clin. Pharmacol. Ther.* **116**, 637–646 (2024).
78. Budde, L. E. et al. Durable responses with mosunetuzumab in relapsed/refractory indolent and aggressive B-cell non-Hodgkin lymphomas: extended follow-up of a phase I/II study. *J. Clin. Oncol.* **42**, 2250–2256 (2024).
79. Dickinson, M. J. et al. Glofitamab for relapsed or refractory diffuse large B-cell lymphoma. *N. Engl. J. Med.* **387**, 2220–2231 (2022).
80. Thieblemont, C. et al. Epcoritamab, a novel, subcutaneous CD3×CD20 bispecific T-cell-engaging antibody, in relapsed or refractory large B-cell lymphoma: dose expansion in a phase I/II trial. *J. Clin. Oncol.* **41**, 2238–2247 (2023).
81. Kim, T. M. et al. Safety and efficacy of odronextamab in patients with relapsed or refractory follicular lymphoma. *Ann. Oncol.* **35**, 1039–1047 (2024).
82. Moreau, P. et al. Teclistamab in relapsed or refractory multiple myeloma. *N. Engl. J. Med.* **387**, 495–505 (2022).
83. Lesokhin, A. M. et al. Elranatamab in relapsed or refractory multiple myeloma: phase 2 MagnetisMM-3 trial results. *Nat. Med.* **29**, 2259–2267 (2023).
84. Avigan, Z. M., Rattu, M. A. & Richter, J. An evaluation of linvoseltamab for treatment of relapsed/refractory multiple myeloma. *Expert Opin. Biol. Ther.* **25**, 221–228 (2025).
85. Chari, A. et al. Talquetamab, a T-cell-redirecting GPRC5D bispecific antibody for multiple myeloma. *N. Engl. J. Med.* **387**, 2232–2244 (2022).
86. Bacac, M. et al. CD20-TCB with obinutuzumab pretreatment as next-generation treatment of hematologic malignancies. *Clin. Cancer Res.* **24**, 4785–4797 (2018).
87. Guedan, S., Ruella, M. & June, C. H. Emerging cellular therapies for cancer. *Annu. Rev. Immunol.* **37**, 145–171 (2019).
88. Haydu, J. E. & Abramson, J. S. The rules of T-cell engagement: current state of CAR T cells and bispecific antibodies in B-cell lymphomas. *Blood Adv.* **8**, 4700–4710 (2024).
89. Swan, D., Madduri, D. & Hocking, J. CAR-T cell therapy in multiple myeloma: current status and future challenges. *Blood Cancer J.* **14**, 206 (2024).
90. Anderson, N. D. et al. Transcriptional signatures associated with persisting CD19 CAR-T cells in children with leukemia. *Nat. Med.* **29**, 1700–1709 (2023).
91. Witzig, T. E. et al. Randomized controlled trial of yttrium-90-labeled ibritumomab tiuxetan radioimmunotherapy versus rituximab immunotherapy for patients with relapsed or refractory low-grade, follicular, or transformed B-cell non-Hodgkin's lymphoma. *J. Clin. Oncol.* **20**, 2453–2463 (2002).
92. Horning, S. J. et al. Efficacy and safety of tositumomab and iodine-131 tositumomab (Bexxar) in B-cell lymphoma, progressive after rituximab. *J. Clin. Oncol.* **23**, 712–719 (2005).
93. Kantarjian, H. M. et al. Inotuzumab ozogamicin versus standard therapy for acute lymphoblastic leukemia. *N. Engl. J. Med.* **375**, 740–753 (2016).
94. Tilly, H. et al. Polatuzumab vedotin in previously untreated diffuse large B-cell lymphoma. *N. Engl. J. Med.* **386**, 351–363 (2022).
95. Caimi, P. F. et al. Loncastuximab tesirine in relapsed or refractory diffuse large B-cell lymphoma (LOTIS-2): a multicentre, open-label, single-arm, phase 2 trial. *Lancet Oncol.* **22**, 790–800 (2021).
96. Tai, Y. T. et al. Novel anti-B-cell maturation antigen antibody-drug conjugate (GSK2857916) selectively induces killing of multiple myeloma. *Blood* **123**, 3128–3138 (2014).
97. Dimopoulos, M. A. et al. Efficacy and safety of single-agent belantamab mafodotin versus pomalidomide plus low-dose dexamethasone in patients with relapsed or refractory multiple myeloma (DREAMM-3): a phase 3, open-label, randomised study. *Lancet Haematol.* **10**, e801–e812 (2023).
98. Kreitman, R. J. et al. Moxetumomab pasudotox in relapsed/refractory hairy cell leukemia. *Leukemia* **32**, 1768–1777 (2018).
99. Khoury, R. et al. Mechanisms of resistance to antibody-drug conjugates. *Int. J. Mol. Sci.* **24**, 9674 (2023).
100. Edwards, J. C. et al. Efficacy of B-cell-targeted therapy with rituximab in patients with rheumatoid arthritis. *N. Engl. J. Med.* **350**, 2572–2581 (2004).
First demonstration of efficacy of rituximab, a chimeric anti-CD20 antibody, in a randomized double-blind placebo-controlled trial of patients with rheumatoid arthritis.
101. Cohen, S. B. et al. Rituximab for rheumatoid arthritis refractory to anti-tumor necrosis factor therapy: results of a multicenter, randomized, double-blind, placebo-controlled, phase III trial evaluating primary efficacy and safety at twenty-four weeks. *Arthritis Rheum.* **54**, 2793–2806 (2006).
102. Joly, P. et al. First-line rituximab combined with short-term prednisone versus prednisone alone for the treatment of pemphigus (Ritux 3): a prospective, multicentre, parallel-group, open-label randomised trial. *Lancet* **389**, 2031–2040 (2017).
103. Jones, R. B. et al. Rituximab versus cyclophosphamide in ANCA-associated renal vasculitis. *N. Engl. J. Med.* **363**, 211–220 (2010).
104. Martin, F. & Chan, A. C. B cell immunobiology in disease: evolving concepts from the clinic. *Annu. Rev. Immunol.* **24**, 467–496 (2006).
105. Merrill, J. T. et al. Efficacy and safety of rituximab in moderately-to-severely active systemic lupus erythematosus: the randomized, double-blind, phase II/III systemic lupus erythematosus evaluation of rituximab trial. *Arthritis Rheum.* **62**, 222–233 (2010).
106. Rovin, B. H. et al. Efficacy and safety of rituximab in patients with active proliferative lupus nephritis: the lupus nephritis assessment with rituximab study. *Arthritis Rheum.* **64**, 1215–1226 (2012).
107. Gomez-Mendez, L. M. et al. Peripheral blood B cell depletion after rituximab and complete response in lupus nephritis. *Clin. J. Am. Soc. Nephrol.* **13**, 1502–1509 (2018).
108. Gong, Q. et al. Importance of cellular microenvironment and circulatory dynamics in B cell immunotherapy. *J. Immunol.* **174**, 817–826 (2005).
109. Ahuja, A. et al. Depletion of B cells in murine lupus: efficacy and resistance. *J. Immunol.* **179**, 3351–3361 (2007).
110. Marinov, D. A. et al. The type II anti-CD20 antibody obinutuzumab (GA101) is more effective than rituximab at depleting B cells and treating disease in a murine lupus model. *Arthritis Rheumatol.* **73**, 826–836 (2021).
111. Furie, R. A. et al. Efficacy and safety of obinutuzumab in active lupus nephritis. *N. Engl. J. Med.* **392**, 1471–1483 (2025).
112. Hauser, S. L. et al. B-cell depletion with rituximab in relapsing-remitting multiple sclerosis. *N. Engl. J. Med.* **358**, 676–688 (2008).
113. Hawker, K. et al. Rituximab in patients with primary progressive multiple sclerosis: results of a randomized double-blind placebo-controlled multicenter trial. *Ann. Neurol.* **66**, 460–471 (2009).

114. Dunn, N. et al. Rituximab in multiple sclerosis: frequency and clinical relevance of anti-drug antibodies. *Mult. Scler.* **24**, 1224–1233 (2018).
115. Wolf, A. B. et al. Rituximab-induced serum sickness in multiple sclerosis patients. *Mult. Scler. Relat. Disord.* **36**, 101402 (2019).
116. Wincup, C. et al. Anti-rituximab antibodies demonstrate neutralizing capacity, associate with lower circulating drug levels and earlier relapse in lupus. *Rheumatology* **62**, 2601–2610 (2023).
117. Hauser, S. L. et al. Ofatumumab versus teriflunomide in multiple sclerosis. *N. Engl. J. Med.* **383**, 546–557 (2020).
118. Hauser, S. L. et al. Ocrelizumab versus interferon beta-1a in relapsing multiple sclerosis. *N. Engl. J. Med.* **376**, 221–234 (2017).
- First phase III demonstration of efficacy of ocrelizumab, a humanized anti-CD20 antibody, compared with interferon β 1a standard of care in relapsing multiple sclerosis.**
119. Montalban, X. et al. Ocrelizumab versus placebo in primary progressive multiple sclerosis. *N. Engl. J. Med.* **376**, 209–220 (2017).
- First phase III demonstration of efficacy of ocrelizumab, a humanized anti-CD20 antibody, in PPMS.**
120. Fischbach, F. et al. CD19-targeted chimeric antigen receptor T cell therapy in two patients with multiple sclerosis. *Med* **5**, 550–558 e552 (2024).
121. Müller, F. et al. CD19 CAR T-cell therapy in autoimmune disease – a case series with follow-up. *N. Engl. J. Med.* **390**, 687–700 (2024).
122. Swain, S. M., Shastry, M. & Hamilton, E. Targeting HER2-positive breast cancer: advances and future directions. *Nat. Rev. Drug Discov.* **22**, 101–126 (2023).
123. Carter, P. et al. Humanization of an anti-p185^{HER2} antibody for human cancer therapy. *Proc. Natl Acad. Sci. USA* **89**, 4285–4289 (1992).
- Humanization of a murine anti-HER2 antibody, 4D5, that led to trastuzumab, the first antibody to be approved for breast cancer.**
124. Hudziak, R. M. et al. p185^{HER2} monoclonal antibody has antiproliferative effects *in vitro* and sensitizes human breast tumor cells to tumor necrosis factor. *Mol. Cell. Biol.* **9**, 1165–1172 (1989).
125. Slamon, D. J. et al. Use of chemotherapy plus a monoclonal antibody against HER2 for metastatic breast cancer that overexpresses HER2. *N. Engl. J. Med.* **344**, 783–792 (2001).
- Demonstration that the humanized anti-HER2 antibody trastuzumab increases the clinical benefit of chemotherapy in metastatic breast cancer that overexpresses HER2.**
126. Jorgensen, J. T. et al. A companion diagnostic with significant clinical impact in treatment of breast and gastric cancer. *Front. Oncol.* **11**, 676939 (2021).
127. Slamon, D. et al. Adjuvant trastuzumab in HER2-positive breast cancer. *N. Engl. J. Med.* **365**, 1273–1283 (2011).
128. Agus, D. B. et al. Targeting ligand-activated ErbB2 signaling inhibits breast and prostate tumor growth. *Cancer Cell* **2**, 127–137 (2002).
129. Shao, Z. et al. Efficacy, safety, and tolerability of pertuzumab, trastuzumab, and docetaxel for patients with early or locally advanced ERBB2-positive breast cancer in Asia: the PEONY phase 3 randomized clinical trial. *JAMA Oncol.* **6**, e193692 (2020).
130. von Minckwitz, G. et al. Adjuvant pertuzumab and trastuzumab in early HER2-positive breast cancer. *N. Engl. J. Med.* **377**, 122–131 (2017).
131. Swain, S. M. et al. Pertuzumab, trastuzumab, and docetaxel in HER2-positive metastatic breast cancer. *N. Engl. J. Med.* **372**, 724–734 (2015).
132. Gao, J. J. et al. FDA approval summary: pertuzumab, trastuzumab, and hyaluronidase-zxxf injection for subcutaneous use in patients with HER2-positive breast cancer. *Clin. Cancer Res.* **27**, 2126–2129 (2021).
133. Hurvitz, S. A. et al. Analysis of Fcy receptor IIIa and IIa polymorphisms: lack of correlation with outcome in trastuzumab-treated breast cancer patients. *Clin. Cancer Res.* **18**, 3478–3486 (2012).
134. Tamura, K. et al. FcyR2A and 3A polymorphisms predict clinical outcome of trastuzumab in both neoadjuvant and metastatic settings in patients with HER2-positive breast cancer. *Ann. Oncol.* **22**, 1302–1307 (2011).
135. Wang, D. S. et al. FcyRIIA and IIIA polymorphisms predict clinical outcome of trastuzumab-treated metastatic gastric cancer. *OncoTargets Ther.* **10**, 5065–5076 (2017).
136. Nordstrom, J. L. et al. Anti-tumor activity and toxicokinetics analysis of MGAH22, an anti-HER2 monoclonal antibody with enhanced Fcy receptor binding properties. *Breast Cancer Res.* **13**, R123 (2011).
137. Fendly, B. M. et al. Characterization of murine monoclonal antibodies reactive to either the human epidermal growth factor receptor or HER2/neu gene product. *Cancer Res.* **50**, 1550–1558 (1990).
138. Rugo, H. S. et al. Efficacy of margetuximab vs trastuzumab in patients with pretreated ERBB2-positive advanced breast cancer: a phase 3 randomized clinical trial. *JAMA Oncol.* **7**, 573–584 (2021).
139. Lewis Phillips, G. D. et al. Targeting HER2-positive breast cancer with trastuzumab-DM1, an antibody-cytotoxic drug conjugate. *Cancer Res.* **68**, 9280–9290 (2008).
140. Verma, S. et al. Trastuzumab emtansine for HER2-positive advanced breast cancer. *N. Engl. J. Med.* **367**, 1783–1791 (2012).
141. von Minckwitz, G. et al. Trastuzumab emtansine for residual invasive HER2-positive breast cancer. *N. Engl. J. Med.* **380**, 617–628 (2019).
142. Ogatani, Y. et al. DS-8201a, a novel HER2-targeting ADC with a novel DNA topoisomerase I inhibitor, demonstrates a promising antitumor efficacy with differentiation from T-DM1. *Clin. Cancer Res.* **22**, 5097–5108 (2016).
143. Ogatani, Y., Hagihara, K., Oitate, M., Naito, H. & Agatsuma, T. Bystander killing effect of DS-8201a, a novel anti-human epidermal growth factor receptor 2 antibody-drug conjugate, in tumors with human epidermal growth factor receptor 2 heterogeneity. *Cancer Sci.* **107**, 1039–1046 (2016).
144. Narayan, P. et al. FDA approval summary: fam-trastuzumab deruxtecan-nxki for the treatment of unresectable or metastatic HER2-positive breast cancer. *Clin. Cancer Res.* **27**, 4478–4485 (2021).
145. Modi, S. et al. Trastuzumab deruxtecan in previously treated HER2-positive breast cancer. *N. Engl. J. Med.* **382**, 610–621 (2020).
146. Modi, S. et al. Trastuzumab deruxtecan in previously treated HER2-low advanced breast cancer. *N. Engl. J. Med.* **387**, 9–20 (2022).
- Landmark clinical study demonstrating the efficacy of the ADC, trastuzumab deruxtecan, in previously treated HER2-low breast cancer.**
147. Prescribing information, ENHERTU[®] (fam-trastuzumab deruxtecan-nxki) for injection, for intravenous use. [fda.gov https://www.accessdata.fda.gov/drugsatfda_docs/label/2025/761139s032s035lbl.pdf](https://www.accessdata.fda.gov/drugsatfda_docs/label/2025/761139s032s035lbl.pdf) (2025).
148. Weisser, N. E. et al. An anti-HER2 biparatopic antibody that induces unique HER2 clustering and complement-dependent cytotoxicity. *Nat. Commun.* **14**, 1394 (2023).
149. Harding, J. J. et al. Zanidatamab for HER2-amplified, unresectable, locally advanced or metastatic biliary tract cancer (HERIZON-BTC-01): a multicentre, single-arm, phase 2b study. *Lancet Oncol.* **24**, 772–782 (2023).
150. Keam, S. J. Zanidatamab: first approval. *Drugs* **85**, 707–714 (2025).
151. Jonna, S. et al. Detection of NRG1 gene fusions in solid tumors. *Clin. Cancer Res.* **25**, 4966–4972 (2019).
152. Werr, L. et al. CD74-NRG1 fusions are oncogenic *in vivo* and induce therapeutically tractable ERBB2:ERBB3 heterodimerization. *Mol. Cancer Ther.* **21**, 821–830 (2022).
153. Schram, A. M. et al. Efficacy of zenocutuzumab in NRG1 fusion-positive cancer. *N. Engl. J. Med.* **392**, 566–576 (2025).
154. Press, M. F., Cordon-Cardo, C. & Slamon, D. J. Expression of the HER-2/neu proto-oncogene in normal human adult and fetal tissues. *Oncogene* **5**, 953–962 (1990).
155. Gordon, L. I. et al. Blockade of the erbB2 receptor induces cardiomyocyte death through mitochondrial and reactive oxygen species-dependent pathways. *J. Biol. Chem.* **284**, 2080–2087 (2009).
156. Seidman, A. et al. Cardiac dysfunction in the trastuzumab clinical trials experience. *J. Clin. Oncol.* **20**, 1215–1221 (2002).
157. Morgan, R. A. et al. Case report of a serious adverse event following the administration of T cells transduced with a chimeric antigen receptor recognizing ERBB2. *Mol. Ther.* **18**, 843–851 (2010).
158. Slaga, D. et al. Avidity-based binding to HER2 results in selective killing of HER2-overexpressing cells by anti-HER2/CD3. *Sci. Transl. Med.* **10**, eaat5775 (2018).
- Avidity engineering of a TCE bispecific antibody (HER2 and CD3) to enhance the selectivity of cytotoxicity towards tumour cells overexpressing HER2.**
159. Oostindie, S. C. et al. Logic-gated antibody pairs that selectively act on cells co-expressing two antigens. *Nat. Biotechnol.* **40**, 1509–1519 (2022).
160. Davis, J. D. et al. Subcutaneous administration of monoclonal antibodies: pharmacology, delivery, immunogenicity, and learnings from applications to clinical development. *Clin. Pharmacol. Ther.* **115**, 422–439 (2024).
161. Lang, J. J. et al. Patient-reported disruptions to cancer care during the COVID-19 pandemic: a national cross-sectional study. *Cancer Med.* **12**, 4773–4785 (2023).
162. Lambert, M. A. & Finlay, W. J. J. in *Orphan Drugs and Rare Disease*. Drug Discovery Series 38 (eds D. C. Pryde & M. J. Palmer) Ch. 14, 401–418 (Royal Society of Chemistry, 2014).
163. Prescribing information, Soliris (eculizumab) for intravenous infusion. [fda.gov https://www.accessdata.fda.gov/drugsatfda_docs/label/2025/125166s448,761108s038lbl.pdf](https://www.accessdata.fda.gov/drugsatfda_docs/label/2025/125166s448,761108s038lbl.pdf) (2025).
164. Sheridan, D. et al. Design and preclinical characterization of ALXN1210: a novel anti-C5 antibody with extended duration of action. *PLoS ONE* **13**, e0195909 (2018).
165. Zalevsky, J. et al. Enhanced antibody half-life improves *in vivo* activity. *Nat. Biotechnol.* **28**, 157–159 (2010).
166. Prescribing information, Ultomiris (ravulizumab-cwvz) injection, for intravenous use. [fda.gov https://www.accessdata.fda.gov/drugsatfda_docs/label/2024/761108s037lbl.pdf](https://www.accessdata.fda.gov/drugsatfda_docs/label/2024/761108s037lbl.pdf) (2024).
167. Viola, M. et al. Subcutaneous delivery of monoclonal antibodies: how do we get there? *J. Control. Release* **286**, 301–314 (2018).
168. Strickley, R. G. & Lambert, W. J. A review of formulations of commercially available antibodies. *J. Pharm. Sci.* **110**, 2590–2608 e2556 (2021).
169. Zarzar, J. et al. High concentration formulation development approaches and considerations. *MAbs* **15**, 2211185 (2023).
170. Sánchez-Félix, M., Burke, M., Chen, H. H., Patterson, C. & Mittal, S. Predicting bioavailability of monoclonal antibodies after subcutaneous administration: open innovation challenge. *Adv. Drug Deliv. Rev.* **167**, 66–77 (2020).
171. Sauna, S. E., Jawa, V., Balu-Iyer, S. & Chirmule, N. Understanding preclinical and clinical immunogenicity risks in novel biotherapeutics development. *Front. Immunol.* **14**, 1151888 (2023).
172. Swanson, S. J. What are clinically significant anti-drug antibodies and why is it important to identify them. *Front. Immunol.* **15**, 1401178 (2024).
173. Homšek, A. et al. Pharmacokinetic characterization, benefits and barriers of subcutaneous administration of monoclonal antibodies in oncology. *J. Oncol. Pharm. Pract.* **29**, 431–440 (2022).

174. Dai, J. et al. Variable domain mutational analysis to probe the molecular mechanisms of high viscosity of an IgG₂ antibody. *MAbs* **16**, 2304282 (2024).
175. Nichols, P. et al. Rational design of viscosity reducing mutants of a monoclonal antibody: hydrophobic versus electrostatic inter-molecular interactions. *MAbs* **7**, 212–230 (2015).
176. Yadav, S. et al. Establishing a link between amino acid sequences and self-associating and viscoelastic behavior of two closely related monoclonal antibodies. *Pharm. Res.* **28**, 1750–1764 (2011).
177. Heisler, J., Kovner, D., Izadi, S., Zarzar, J. & Carter, P. J. Modulation of the high concentration viscosity of IgG₂ antibodies using clinically validated Fc mutations. *MAbs* **16**, 2379560 (2024).
178. Igawa, T., Haraya, K. & Hattori, K. Sweeping antibody as a novel therapeutic antibody modality capable of eliminating soluble antigens from circulation. *Immunol. Rev.* **270**, 132–151 (2016).
179. Igawa, T. et al. Engineered monoclonal antibody with novel antigen-sweeping activity *in vivo*. *PLoS ONE* **8**, e63236 (2013).
180. Igawa, T. et al. Antibody recycling by engineered pH-dependent antigen binding improves the duration of antigen neutralization. *Nat. Biotechnol.* **28**, 1203–1207 (2010).
First demonstration that engineering an antibody for pH-dependent antigen binding could extend its plasma life and duration of antigen neutralization in vivo.
181. Sampei, Z. et al. Antibody engineering to generate SKY59, a long-acting anti-C5 recycling antibody. *PLoS ONE* **13**, e0209509 (2018).
182. Fukuzawa, T. et al. Long lasting neutralization of C5 by SKY59, a novel recycling antibody, is a potential therapy for complement-mediated diseases. *Sci. Rep.* **7**, 1080 (2017).
183. Callaway, E. Chemistry Nobel goes to developers of AlphaFold AI that predicts protein structures. *Nature* **634**, 525–526 (2024).
184. Jumper, J. et al. Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, 583–589 (2021).
First release of AlphaFold, an artificial intelligence tool that predicts protein structures from amino acid sequences with previously unprecedented speed and accuracy, with great potential to accelerate basic research and drug discovery.
185. Vazquez Torres, S. et al. De novo design of high-affinity binders of bioactive helical peptides. *Nature* **626**, 435–442 (2024).
186. Ichikawa, D. M. et al. A universal deep-learning model for zinc finger design enables transcription factor reprogramming. *Nat. Biotechnol.* **41**, 1117–1129 (2023).
187. Hie, B. L. et al. Efficient evolution of human antibodies from general protein language models. *Nat. Biotechnol.* **42**, 275–283 (2024).
188. Frey, N. C. et al. Lab-in-the-loop therapeutic antibody design with deep learning. Preprint at *bioRxiv* <https://doi.org/10.1101/2025.02.19.639050> (2025).
189. Li, L. et al. Machine learning optimization of candidate antibody yields highly diverse sub-nanomolar affinity antibody libraries. *Nat. Commun.* **14**, 3454 (2023).
190. Harvey, E. P. et al. An *in silico* method to assess antibody fragment polyreactivity. *Nat. Commun.* **13**, 7554 (2022).
191. Sweet-Jones, J. & Martin, A. C. R. An antibody developability triaging pipeline exploiting protein language models. *MAbs* **17**, 2472009 (2025).
192. Kalejaye, L. A. et al. Accelerating high-concentration monoclonal antibody development with large-scale viscosity data and ensemble deep learning. *MAbs* **17**, 2483944 (2025).
193. Liang, W. C. et al. Structure- and machine learning-guided engineering demonstrate that a non-canonical disulfide in an anti-PP-1 rabbit antibody does not impede antibody developability. *MAbs* **16**, 2309685 (2024).
194. Wang, H., Hao, X., He, Y. & Fan, L. AblImmPred: an immunogenicity prediction method for therapeutic antibodies using AntiBERTy-based sequence features. *PLoS ONE* **19**, e0296737 (2024).
195. Bennett, N. R. et al. Atomically accurate de novo design of single-domain antibodies with RFdiffusion. Preprint at *bioRxiv* <https://doi.org/10.1101/2024.03.14.585103> (2025).
196. Chungyoun, M. & Gray, J. J. AI models for protein design are driving antibody engineering. *Curr. Opin. Biomed. Eng.* **28**, 100473 (2023).
197. Zheng, J., Wang, Y., Liang, Q., Cui, L. & Wang, L. The application of machine learning on antibody discovery and optimization. *Molecules* **29**, 5923 (2024).
198. Desnoyers, L. R. et al. Tumor-specific activation of an EGFR-targeting antibody enhances therapeutic index. *Sci. Transl. Med.* **5**, 207ra144 (2013).
199. Chang, H. W. et al. Generating tumor-selective conditionally active biologic anti-CTLA4 antibodies via protein-associated chemical switches. *Proc. Natl Acad. Sci. USA* **118**, e2020606118 (2021).
200. Mimoto, F. et al. Exploitation of elevated extracellular ATP to specifically direct antibody to tumor microenvironment. *Cell Rep.* **33**, 108542 (2020).
201. Kamata-Sakurai, M. et al. Antibody to CD137 activated by extracellular adenosine triphosphate is tumor selective and broadly effective *in vivo* without systemic immune activation. *Cancer Discov.* **11**, 158–175 (2021).
202. Hiraniwa, N. et al. Calcium-dependent antigen binding as a novel modality for antibody recycling by endosomal antigen dissociation. *MAbs* **8**, 65–73 (2016).
203. Chu, T. H., Patz, E. F. Jr. & Ackerman, M. E. Coming together at the hinges: therapeutic prospects of IgG3. *MAbs* **13**, 1882028 (2021).
204. Keyt, B. A., Baliga, R., Sinclair, A. M., Carroll, S. F. & Peterson, M. S. Structure, function, and therapeutic use of IgM antibodies. *Antibodies* **9**, 53 (2020).
205. Buchner, J., Sitia, R. & Svilenov, H. L. Understanding IgM structure and biology to engineer new antibody therapeutics. *BioDrugs* **39**, 347–357 (2025).
206. Bohländer, F. A new hope? Possibilities of therapeutic IgA antibodies in the treatment of inflammatory lung diseases. *Front. Immunol.* **14**, 1127339 (2023).
207. Candelaria, P. V., Nava, M., Daniels-Wells, T. R. & Penichet, M. L. A fully human IgE specific for CD38 as a potential therapy for multiple myeloma. *Cancers* **15**, 4533 (2023).
208. Chauhan, J. et al. IgE antibodies against cancer: efficacy and safety. *Antibodies* **9**, 55 (2020).
209. Ku, Z. et al. Nasal delivery of an IgM offers broad protection from SARS-CoV-2 variants. *Nature* **595**, 718–723 (2021).
210. Marks, L. The birth pangs of monoclonal antibody therapeutics: the failure and legacy of Centoxin. *MAbs* **4**, 403–412 (2012).
211. de Jong, R. N. et al. A novel platform for the potentiation of therapeutic antibodies based on antigen-dependent formation of IgG hexamers at the cell surface. *PLoS Biol.* **14**, e1002344 (2016).
First demonstration of antigen-dependent IgG1 hexamerization (HexaBody technology), which was enabling for several such antibodies that have reached clinical trials.
212. Grandits, M. et al. Hybrid IgE-IgG1 antibodies (IgEG): a new antibody class that combines IgE and IgG functionality. *MAbs* **17**, 2502673 (2025).
213. Carter, P. J. & Lazar, G. A. Next generation antibody drugs: pursuit of the ‘high-hanging fruit’. *Nat. Rev. Drug Discov.* **17**, 197–223 (2018).
214. Lim, S. H., Beers, S. A., Al-Shamkhani, A. & Cragg, M. S. Agonist antibodies for cancer immunotherapy: history, hopes, and challenges. *Clin. Cancer Res.* **30**, 1712–1723 (2024).
215. Jhaji, H. S., Lwo, T. S., Yao, E. L. & Tessier, P. M. Unlocking the potential of agonist antibodies for treating cancer using antibody engineering. *Trends Mol. Med.* **29**, 48–60 (2023).
216. Yen, M. et al. Facile discovery of surrogate cytokine agonists. *Cell* **185**, 1414–1430.e1419 (2022).
217. Pekar, L., Krah, S. & Zielonka, S. Taming the beast: engineering strategies and biomedical potential of antibody-based cytokine mimetics. *Expert Opin. Biol. Ther.* **24**, 115–118 (2024).
218. Terstappen, G. C., Meyer, A. H., Bell, R. D. & Zhang, W. Strategies for delivering therapeutics across the blood–brain barrier. *Nat. Rev. Drug Discov.* **20**, 362–383 (2021).
219. Yu, Y. J. et al. Boosting brain uptake of a therapeutic antibody by reducing its affinity for a transcytosis target. *Sci. Transl. Med.* **3**, 84ra44 (2011).
220. Grimm, H. P. et al. Delivery of the Brainshuttle amyloid-beta antibody fusion trontinemab to non-human primate brain and projected efficacious dose regimens in humans. *MAbs* **15**, 2261509 (2023).
221. Okuyama, T. et al. A phase 2/3 trial of pabinafusp alfa, IDS fused with anti-human transferrin receptor antibody, targeting neurodegeneration in MPS-II. *Mol. Ther.* **29**, 671–679 (2021).
The fusion protein abinafusp alfa provides the first clinical proof of concept of using transferrin-mediated transcytosis to enhance delivery of a protein therapeutic into the central nervous system.
222. Deng, B. et al. Oral nanomedicine: challenges and opportunities. *Adv. Mater.* **36**, e2306081 (2024).
223. Nicze, M. et al. The current and promising oral delivery methods for protein- and peptide-based drugs. *Int. J. Mol. Sci.* **25**, 815 (2024).
224. Abramson, A. et al. An ingestible self-orienting system for oral delivery of macromolecules. *Science* **363**, 611–615 (2019).
Preclinical proof of concept of oral delivery of a protein therapeutic (insulin) into systemic circulation using a sophisticated ingestible medical device.
225. Abramson, A. et al. Oral delivery of systemic monoclonal antibodies, peptides and small molecules using gastric auto-injectors. *Nat. Biotechnol.* **40**, 103–109 (2022).
226. Ota, N. et al. Engineering a protease-stable, oral single-domain antibody to inhibit IL-23 signaling. *Proc. Natl Acad. Sci. USA* **122**, e2501635122 (2025).
227. Harris, C. T. & Cohen, S. Reducing immunogenicity by design: approaches to minimize immunogenicity of monoclonal antibodies. *BioDrugs* **38**, 205–226 (2024).
228. Ridker, P. M. et al. Lipid-reduction variability and antidrug-antibody formation with bococizumab. *N. Engl. J. Med.* **376**, 1517–1526 (2017).
229. Kearns, J. D. et al. A root cause analysis to identify the mechanistic drivers of immunogenicity against the anti-VEGF biotherapeutic brolocizumab. *Sci. Transl. Med.* **15**, eabq5068 (2023).
230. Dyson, M. R. et al. Beyond affinity: selection of antibody variants with optimal biophysical properties and reduced immunogenicity from mammalian display libraries. *MAbs* **12**, 1829335 (2020).
Demonstration that improving the biophysical properties of an antibody (bococizumab) correlates with reduced immunogenicity risk in nonclinical assays.
231. Schmitt, C. et al. Low immunogenicity of emicizumab in persons with haemophilia A. *Haemophilia* **27**, 984–992 (2021).
232. Sampei, Z. et al. Identification and multidimensional optimization of an asymmetric bispecific IgG antibody mimicking the function of factor VIII cofactor activity. *PLoS ONE* **8**, e57479 (2013).
Elegant antibody engineering that led to emicizumab, a bispecific antibody (to factor IXa and factor X) with a common light chain approved for treatment of haemophilia A.
233. Ghosh, I., Gutka, H., Krause, M. E., Clemens, R. & Kashi, R. S. A systematic review of commercial high concentration antibody drug products approved in the US: formulation composition, dosage form design and primary packaging considerations. *MAbs* **15**, 2205540 (2023).

234. Dall'Acqua, W. F. et al. Increasing the affinity of a human IgG1 for the neonatal Fc receptor: biological consequences. *J. Immunol.* **169**, 5171–5180 (2002).
Antibody engineering that led to the YTE (M252Y:S254T:T256E) triple Fc mutant, which has been used to extend the pharmacokinetic half-life of several approved antibody therapeutics, including nirsevimab, netakimab, amubarvimab and romlusevimab.
235. Gaudinski, M. R. et al. Safety and pharmacokinetics of the Fc-modified HIV-1 human monoclonal antibody VRC01LS: a phase 1 open-label clinical trial in healthy adults. *PLoS Med.* **15**, e1002493 (2018).
236. Robbie, G. J. et al. A novel investigational Fc-modified humanized monoclonal antibody, motavizumab-YTE, has an extended half-life in healthy adults. *Antimicrob. Agents Chemother.* **57**, 6147–6153 (2013).
First demonstration of antibody half-life extension in human by using Fc mutants (YTE, M252Y:S254T:T256E).
237. Zou, P. Predicting human bioavailability of subcutaneously administered fusion proteins and monoclonal antibodies using human intravenous clearance or antibody isoelectric point. *AAPS J.* **25**, 31 (2023).
238. Mulvey, A., Trueb, L., Coukos, G. & Arber, C. Novel strategies to manage CAR-T cell toxicity. *Nat. Rev. Drug Discov.* **24**, 379–397 (2025).
239. Oldham, R. K. Monoclonal antibodies in cancer therapy. *J. Clin. Oncol.* **1**, 582–590 (1983).
240. Ritz, J. & Schlossman, S. F. Utilization of monoclonal antibodies in the treatment of leukemia and lymphoma. *Blood* **59**, 1–11 (1982).
241. Saleh, M. N. et al. A phase II trial of murine monoclonal antibody 17-1A and interferon- γ : clinical and immunological data. *Cancer Immunol. Immunother.* **32**, 185–190 (1990).
242. Shawler, D. L., Bartholomew, R. M., Smith, L. M. & Dillman, R. O. Human immune response to multiple injections of murine monoclonal IgG. *J. Immunol.* **135**, 1530–1535 (1985).
243. Münz, M. et al. Side-by-side analysis of five clinically tested anti-EpCAM monoclonal antibodies. *Cancer Cell Int.* **10**, 44 (2010).
244. Goodman, G. E., Beaumier, P., Hellström, I., Fernyhough, B. & Hellström, K. E. Pilot trial of murine monoclonal antibodies in patients with advanced melanoma. *J. Clin. Oncol.* **3**, 340–352 (1985).
245. Vidarsson, G., Dekkers, G. & Rispens, T. IgG subclasses and allotypes: from structure to effector functions. *Front. Immunol.* **5**, 520 (2014).
246. Ober, R. J., Radu, C. G., Ghetie, V. & Ward, E. S. Differences in promiscuity for antibody–FcRn interactions across species: implications for therapeutic antibodies. *Int. Immunol.* **13**, 1551–1559 (2001).
247. Foote, J. & Winter, G. Antibody framework residues affecting the conformation of the hypervariable loops. *J. Mol. Biol.* **224**, 487–499 (1992).
248. Rothe, C. et al. The human combinatorial antibody library HuCAL GOLD combines diversification of all six CDRs according to the natural immune system with a novel display method for efficient selection of high-affinity antibodies. *J. Mol. Biol.* **376**, 1182–1200 (2008).
249. Hao, Y. et al. Synthetic integrin antibodies discovered by yeast display reveal α V subunit pairing preferences with β subunits. *MAbs* **16**, 2365891 (2024).
250. Porebski, B. T. et al. Rapid discovery of high-affinity antibodies via massively parallel sequencing, ribosome display and affinity screening. *Nat. Biomed. Eng.* **8**, 214–232 (2024).
251. Lee, E. C. et al. Complete humanization of the mouse immunoglobulin loci enables efficient therapeutic antibody discovery. *Nat. Biotechnol.* **32**, 356–363 (2014).
252. Murphy, A. J. et al. Mice with megabase humanization of their immunoglobulin genes generate antibodies as efficiently as normal mice. *Proc. Natl Acad. Sci. USA* **111**, 5153–5158 (2014).
253. Osborn, M. J. et al. High-affinity IgG antibodies develop naturally in Ig-knockout rats carrying germline human IgH/Ig κ /Ig λ loci bearing the rat C μ region. *J. Immunol.* **190**, 1481–1490 (2013).
254. Ouisse, L. H. et al. Antigen-specific single B cell sorting and expression-cloning from immunoglobulin humanized rats: a rapid and versatile method for the generation of high affinity and discriminative human monoclonal antibodies. *BMC Biotechnol.* **17**, 3 (2017).
255. Ros, F. et al. Rabbits transgenic for human IgG genes recapitulating rabbit B-cell biology to generate human antibodies of high specificity and affinity. *MAbs* **12**, 1846900 (2020).
256. Ching, K. H. et al. Chickens with humanized immunoglobulin genes generate antibodies with high affinity and broad epitope coverage to conserved targets. *MAbs* **10**, 71–80 (2018).
257. Boyd, S. D. & Crowe, J. E. Jr. Deep sequencing and human antibody repertoire analysis. *Curr. Opin. Immunol.* **40**, 103–109 (2016).
258. Kelley, B. The history and potential future of monoclonal antibody therapeutics development and manufacturing in four eras. *MAbs* **16**, 2373330 (2024).
259. Carter, P. J. Introduction to current and future protein therapeutics: a protein engineering perspective. *Exp. Cell. Res.* **317**, 1261–1269 (2011).
260. Hummel, J. et al. Modeling the downstream processing of monoclonal antibodies reveals cost advantages for continuous methods for a broad range of manufacturing scales. *Biotechnol. J.* **14**, e1700665 (2018).
261. Barnard, G. C., Zhou, M., Shen, A., Yuk, I. H. & Laird, M. W. Utilizing targeted integration CHO pools to potentially accelerate the GMP manufacturing of monoclonal and bispecific antibodies. *Biotechnol. Prog.* **40**, e3399 (2024).
262. Glinšek, K., Bozovičar, K. & Bratkovič, T. CRISPR technologies in Chinese hamster ovary cell line engineering. *Int. J. Mol. Sci.* **24**, 8144 (2023).
263. Ritacco, F. V., Wu, Y. & Khetan, A. Cell culture media for recombinant protein expression in Chinese hamster ovary (CHO) cells: history, key components, and optimization strategies. *Biotechnol. Prog.* **34**, 1407–1426 (2018).
264. Xu, W. J., Lin, Y., Mi, C. L., Pang, J. Y. & Wang, T. Y. Progress in fed-batch culture for recombinant protein production in CHO cells. *Appl. Microbiol. Biotechnol.* **107**, 1063–1075 (2023).
265. MacDonald, M. A. et al. Perfusion culture of Chinese hamster ovary cells for bioprocessing applications. *Crit. Rev. Biotechnol.* **42**, 1099–1115 (2022).
266. Kumar, A., Udugama, I. A., Gargalo, C. L. & Gernaey, K. V. Why is batch processing still dominating the biologics landscape? Towards an integrated continuous bioprocessing alternative. *Processes* **8**, 1641 (2020).
267. Dorival-García, N. et al. Large-scale assessment of extractables and leachables in single-use bags for biomanufacturing. *Anal. Chem.* **90**, 9006–9015 (2018).
268. Amasawa, E., Kuroda, H., Okamura, K., Badr, S. & Sugiyama, H. Cost-benefit analysis of monoclonal antibody cultivation scenarios in terms of life cycle environmental impact and operating cost. *ACS Sustain. Chem. Eng.* **9**, 14012–14021 (2021).
269. Samaras, J. J., Micheletti, M. & Ding, W. Transformation of biopharmaceutical manufacturing through single-use technologies: current state, remaining challenges, and future development. *Annu. Rev. Chem. Biomol. Eng.* **13**, 73–97 (2022).

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A.C.C., G.D.M. and P.J.C. researched data for this article and wrote it.

Competing interests

The authors are employees of Genentech, which develops and commercializes therapeutics, including antibodies and other proteins.

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